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Increasing electricity exports through delivery-informed allocation in distribution networks[†]

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When distribution networks are constrained, how export access is shared affects how much locally generated electricity reaches the grid. In request-based allocation, access is assigned before participants know how much they can export. Unused export permission can block delivery elsewhere, but a shortfall alone cannot establish that loss. Automatically tightening future access can also prevent useful delivery when output recovers. We model how participants choose requests before their available output is known. The analysis identifies when a larger request improves the participant's expected payoff but reduces total delivery. We propose externality-conditioned counterfactual feedback (ECF) to update future request ceilings using completed operating records. After each interval, ECF recalculates the allocation for that interval, capping requests from participants with unused awards at their completed delivery. The calculation retains the interval's loads, network topology, limits and allocation rule. It uses post-event estimates of available export to calculate what each participant could have delivered. Future ceilings tighten for the unused-award group only if the alternative releases access for others, increases their delivery and total delivery, and passes alternating-current (AC) power-flow checks. Otherwise, ceilings recover toward registered capacity. Controlled medium-voltage simulations test five allocation rules. With a moderate update weight and no shortfall charge, ECF delivers more electricity on average than spreading the same total request reduction proportionally across participants. In a separate 80-day Australian simulation using public photovoltaic and load profiles, ECF increases modeled delivery by 0.59% relative to either no feedback or this equal-total control. Total authorization rises while its unused portion falls relative to no feedback. A control that applies the same ceiling update after every unused award leaves less permission unused than ECF, yet ECF delivers 2.76% more electricity. Giving the latest interval full weight instead reduces total delivery below the no-feedback level. These findings distinguish increasing delivered electricity from merely reducing unused permission. They show how completed operating records can improve future export allocation without changing network limits or replacing the allocation rule.

Broader context

The growth of distributed generation makes better use of existing network capacity important alongside network expansion. Flexible export arrangements in South Australia, California, and the EU manage generation within network limits^{1–3}; China's distributed photovoltaic requirements also emphasize connection capacity and controllability⁴. These developments make access to

constrained networks an ongoing operating concern. Yet permission to export and electricity delivered are different outcomes. Where participants request access before output is known, an unused award can prevent another connection from supplying electricity that was available. Reducing requests after every shortfall is not enough: it can also remove access needed when output recovers. The challenge is to learn from unused awards without treating all of them as lost opportunities. This study uses completed operating records to identify when an alternative allocation could have delivered more electricity, then uses that comparison to guide later requests. Existing network limits, allocation rules, and settlement arrangements remain in place. The resulting operating principle is to judge an adjustment by the delivery it enables, not simply by the unused permission it removes. It requires a reliable record of available export, because metered shortfall alone cannot show what others could have delivered.

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[†] Supporting Information (SI), code, calculated results, and figure data are deposited at <https://github.com/MarxHuang/ecf-export-access>. Access is currently restricted to author review; see Data availability.

1 Introduction

1.1 Background and Motivation

Distributed generation, including photovoltaic (PV) generation, can supply the wider power system only if local networks can carry the electricity not consumed on-site. Voltage and equipment-loading limits constrain simultaneous export at some locations and times. Operating envelopes translate these constraints into connection-level bounds^{5–7}, which increasingly form part of flexible export and connection arrangements^{1–4}. When several connections share a binding limit, assigning access to one can restrict another. How access is shared then affects not only permission to export, but also how much available electricity reaches the network.

Request-based allocation introduces a timing problem. Participants ask for access before their available export is fully known, and the allocation rule divides access using those requests. A large request protects delivery if output is high, but can leave part of the resulting award unused if output is low. Such a request can reflect uncertainty, private information, or a rational preference to preserve access for a high-output outcome^{8–10}. Its value to the participant need not coincide with its effect on other connections.

Fig. 1 shows why an unused award alone cannot reveal lost delivery. Two connections share a binding network constraint. In (a), both use their awards fully. In (b) and (c), connection B leaves the same portion unused, but the consequences differ. Giving A more access would not increase delivery in (b), because A has no additional electricity available. In (c), A could use that access, so an alternative allocation under the same conditions would allow more delivery. An award and its meter reading reveal the shortfall, but cannot distinguish these two situations.

Neither charging for a shortfall nor reducing all requests answers this allocation question on its own. A shortfall charge changes the participant's incentive to request access, whereas the delivery lost elsewhere depends on who could have used it. Broad reductions can avoid unused awards but also remove access needed during high-output intervals. The operational challenge is therefore to use the information revealed after delivery to improve later allocation, while retaining useful access under uncertainty.

This question extends the closed-loop use of operating information in cyber-physical power systems^{11,12} to the sharing of constrained export access. Completed records reveal more than whether a participant used its award: together with available-export information, they can show whether a different allocation would have allowed others to deliver more. The question is how to use that distinction in later requests without discarding the protection that advance access provides.

1.2 Related Work

Operating-envelope research establishes how much export a network can support. Optimal-power-flow formulations convert voltage and equipment limits into connection bounds^{5,6}. Robust formulations protect these bounds against uncertain feeder conditions^{13,14}. Regulatory guidance and demonstration specifications show how flexible limits can be communicated and admin-

istered^{15–17}. Market and capacity studies incorporate the bounds into secure exchange, bidding, and capacity assessment^{7,18–22}.

Within network limits, allocation methods divide access using weighted proportional, equitable, or bargaining objectives^{23–26}. Curtailment studies further show that sharing outcomes depend on the chosen fairness measure and allocation objective^{27–30}. Together, these approaches make export allocation sensitive to network limits and to the chosen sharing objective. Feasibility, however, does not ensure that the resulting permission will be used: actual output becomes known after an award based on requests and forecasts has been issued.

Allocated limits can also be exchanged before delivery. SecuLEx lets customers trade portions of their operating envelopes before the operating interval, subject to network-security checks³¹. This adjusts already-issued limits as forecasts or preferences change.

Operating records can also influence subsequent limits. Local-voltage methods use measurements to calculate operating envelopes³². Moring et al. formulate fairness across multiple intervals using joint allocation or, in a separate method, allocation objectives weighted by past curtailment³³. A recent preprint combines accumulated delivery ratios, pricing, and constrained matching in a stateful coordination layer³⁴. These approaches already connect measured or historical conditions to allocation. Our question concerns a different use of the completed record: whether an alternative allocation would have allowed other participants to deliver electricity that was available at that time.

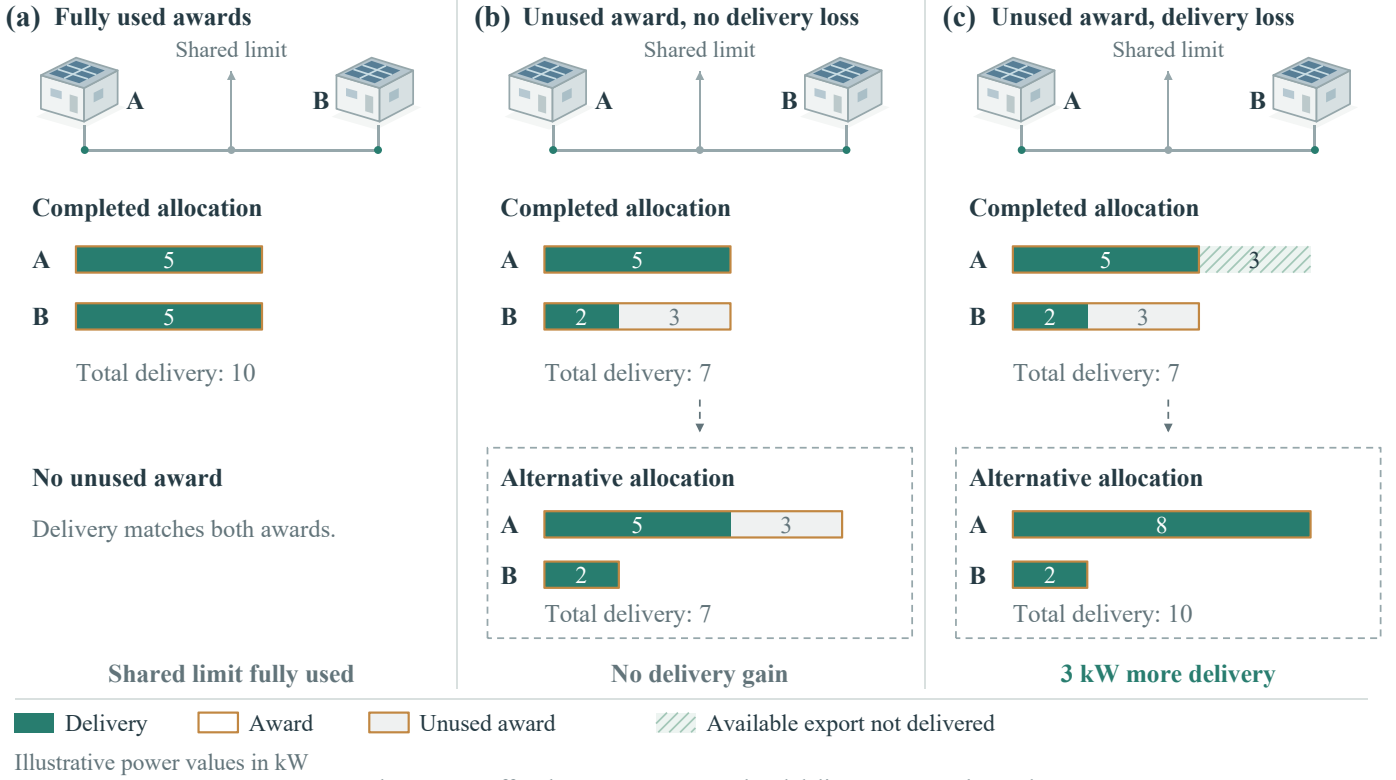
Participant-choice models explain why those advance requests matter. Local energy and flexibility-market studies connect network constraints with bids, prices, payments, and clearing outcomes^{35–37}. Game and mechanism models derive participant choices from incentives rather than treating them as fixed. Demand-response research shows how reported baselines respond to private incentives, uncertain events, and later meter data^{9,10,38}. Distribution-level flexibility procurement also represents uncertain resource availability and unreliable delivery^{8,39–41}. These studies connect requests to incentives and assign consequences to unreliable delivery. They motivate treating larger requests as choices under uncertainty rather than as arbitrary reporting errors. They also clarify the distinction needed here: a charge based on one participant's shortfall and the delivery forgone by other participants need not have the same magnitude or depend on the same information.

These approaches provide network-feasible allocation, incentives, and several ways to use operating history. The distinction examined here is not whether history should matter, but what a completed shortfall justifies changing. Awards and meter readings identify unused permission. Available-export information is also needed to determine whether someone else could have used it. We use that comparison to guide subsequent requests while retaining the underlying allocation rule.

1.3 Contributions

We first identify when protecting a participant's own export reduces total delivery. A single uncertain-output model generates both request choice and later delivery, linking the incentive for a larger request to its effect on other connections. The result-

Two connections share the same network constraint



Settlement can affect later requests. Completed delivery stays unchanged.

Fig. 1 When unused export permission displaces delivery. Two connections share a binding network constraint. (a) Both awards are fully used. (b) B leaves part of its award unused, but A has no additional available export. An alternative allocation increases A's award without increasing delivery. (c) B leaves the same portion unused while A has available export that its award prevents it from delivering. The alternative allocation increases total delivery. Teal fills show delivery, amber outlines show awards, and hatching shows available export not delivered. Numbers within each segment give its power in kW. Dashed frames show calculations for the completed interval. Settlement can affect later requests but cannot restore that interval's lost delivery. The illustrative values use a shared 10 kW authorization bound and initial awards of 5 kW each; they are not case-study results.

ing request–access–delivery externality distinguishes productive larger requests from those that increase the requesting group's payoff and unused award while lowering total delivery. The analysis gives conditions for this divergence and evaluates finite changes when active network constraints change.

We then introduce externality-conditioned counterfactual feedback (ECF), which uses recoverable delivery rather than short-fall alone to select later request updates. It compares the completed allocation with a physically feasible alternative and tightens future ceilings only when that alternative recovers delivery elsewhere and increases the total. A bounded update preserves the existing allocation rule and limits the response to the latest record. The comparison qualifies an unused-award group; it does not assign individual responsibility or require private payoff information.

Two matched controls support these contributions. Equal-total reduction isolates where requests are reduced, while the same update without replay qualification isolates the value of checking recoverable delivery. Controlled comparisons connect these effects to request incentives and allocation rules; public-data simulations examine dynamic response, information quality, and spatial variation. Section 2 explains the operating sequence, Results

develops the findings, and Methods specifies the model. The *Supporting Information*[†] supplies the full derivations and supporting analyses.

2 From unused awards to later request updates

The contrast in Fig. 1(b,c) motivates a comparison of allocations for the completed interval. ECF uses this comparison to decide whether to tighten later request ceilings. A preset feedback gain controls the size of each update.

For each interval t , participant i submits request $r_i(t)$ before its available export is realized. The network-constrained allocation rule grants an export award $x_i(t)$, which is the access authorized for that interval. Available export $a_i(t)$ is realized during the interval, and delivery is

$$y_i(t) = \min\{x_i(t), a_i(t)\}. \quad (1)$$

The award $x_i(t)$ is permission to export; delivery $y_i(t)$ is the electricity actually supplied. We use “award”, “authorized access”, and “authorization” for this same permission. Available export $a_i(t)$ is what the connection could supply before that permission limits delivery. Thus an unused award, $x_i(t) - y_i(t)$, differs from

Table 1 Quantities used in allocation and post-event replay. Power quantities are in MW; t is the allocation interval, written as day and half-hour (d, τ) in the Australian study.

Quantity	Meaning and time of use
$r_i(t)$	Submitted request, made before available export is known.
$r_i^{\text{cap}}(t), r_i^{\text{adm}}(t)$	The same request after the connection bound and then the current feedback ceiling are applied. Allocation uses the admitted request.
$x_i(t), y_i(t)$	Awarded access and completed delivery, respectively. Their difference is the unused award.
$a_i(t)$	Available export before the award limits delivery; realized during the interval.
$a_i^{\text{prof}}(t)$	Australian available-export proxy constructed from public PV and load profiles. It generates modeled delivery.
$a_i^{\text{tel}}(t)$	Post-event record supplied to the replay. It equals $a_i(t)$ in the controlled study and $a_i^{\text{prof}}(t)$ in the primary public replay.
$C_i, \psi_i(t)$	Fixed connection bound and current request ceiling. ECF updates only the latter.
Superscript rpl	An allocation or delivery calculated for the completed-interval comparison.

available electricity that cannot be delivered. Table 1 collects the quantities needed to follow the sequence.

After an interval, metering can reveal an unused award. To determine whether others could have used it, the coordinator also needs a record of available export at each connection. In operation, this record would come from authenticated inverter or plant-controller telemetry, or an independently validated portfolio calculation. The controlled study supplies exact simulated availability. The Australian study constructs availability from public PV and load profiles, so its delivered export is a model outcome. Table 1 distinguishes that proxy from the record supplied to the replay; sensitivity tests perturb the latter while retaining the former.

The *same-state replay* is an allocation calculation for that completed interval. It lowers the unused-award group's replay requests to completed delivery and solves the allocation again. Loads, fixed injections, topology, network limits, rule parameters, and available export remain fixed; awards and power-flow variables are recalculated. The comparison is valid when changing access does not change the underlying available export. Methods states this assumption and treats bounded record error. In the controlled study, a conic network approximation allocates access, and nonlinear radial alternating-current (AC) power flow checks the completed and replayed states.

ECF tightens a later request ceiling only when the replay releases access to participants outside the unused-award group, increases their delivery, increases total delivery, and passes the specified AC checks. If all four conditions hold, ceilings for the unused-award group move toward completed delivery; other ceilings move toward their fixed connection bounds. If any condition fails, all ceilings move toward those bounds. Fig. 2 separates the completed operation, the post-event comparison, and the update for the next interval. The calculation changes future request ceilings, not the completed allocation or delivery.

The outside access and delivery changes describe how recovery occurs. With an exact availability record and the same numerical threshold, increased total delivery implies both outside gains

Table 2 ECF and comparison mechanisms. C denotes the controlled study and A the Australian replay. Network data, allocation rule, and available-export trajectory remain fixed within each comparison.

Mechanism	What changes	Question tested
No feedback (C, A)	Requests are capped only by the fixed connection bound, without an adaptive ceiling.	Baseline delivery without a post-event update.
Shortfall-cost benchmark (C)	A modeled cost of non-delivery changes the participant's request choice before allocation.	How a stronger incentive changes requests and delivery.
Mismatch-only feedback (C)	A smoothed delivery-to-award ratio sets the next ceiling, without replay.	Whether own shortfall alone is a useful basis for updating access.
ECF without replay qualification (A)	The same ECF update applies after every unused award, with the same gain and capacity-recovery target.	Whether checking recoverable delivery adds value beyond the update formula.
Equal-total control (C, A)	One proportional factor removes the same total request as ECF at each interval.	Whether the distribution of request reduction matters.
Previous-day persistence (A)	The request uses previous-day available export with no forecast increment or replay.	Whether conservative requests save unused access at the expense of delivery.
ECF (C, A)	When all four replay conditions hold, the unused-award group's later ceilings move toward completed delivery. Other ceilings recover.	Whether selective use of completed records improves later delivery.

(Section S3.1 of the *Supporting Information*[†]). They are therefore not independent filters in that model. The implementation retains all three comparisons and checks AC feasibility separately; Methods also specifies the treatment of uncertain records.

Two comparisons serve different purposes in this paper. The economic analysis compares larger requests with forecast-based reference requests. The term *request-access-delivery externality* denotes a higher modeled group payoff and a larger unused award accompanied by lower total delivery; it is not a measure of total social welfare. ECF compares a completed allocation with its replay and uses operating records to decide whether later ceilings should tighten. It does not need the private payoff. The frequency of classified externality cases and the frequency of ECF activation therefore measure different events.

The update in Fig. 2 moves the current ceiling $\psi_i(t)$ toward the selected target $\psi_i^{\text{tar}}(t)$. The gain β_{ECF} controls this move; the fixed connection bound C_i does not change. Methods gives the exact equations.

The comparison mechanisms test the delivery consequences of different updates (Table 2). Mismatch-only feedback uses a participant's own delivery-to-award ratio without checking delivery elsewhere. The equal-total control removes exactly as much total request as ECF in each interval but uses a common proportional reduction for everyone. It therefore distinguishes the amount of request removed from its distribution across participants. Its matched total comes from ECF, making it an explanatory control rather than an independently operated policy. Previous-day persistence provides a further public-data comparison with more conservative requests. Table 3 links the resulting questions to the experiments; Methods and the *Supporting Information*[†] give their construction.

3 Results and discussion

Matching the total request reduction does not match delivered electricity. At zero shortfall cost and gain 0.5, ECF delivers 24.8 kW more than the equal-total control in the controlled study.

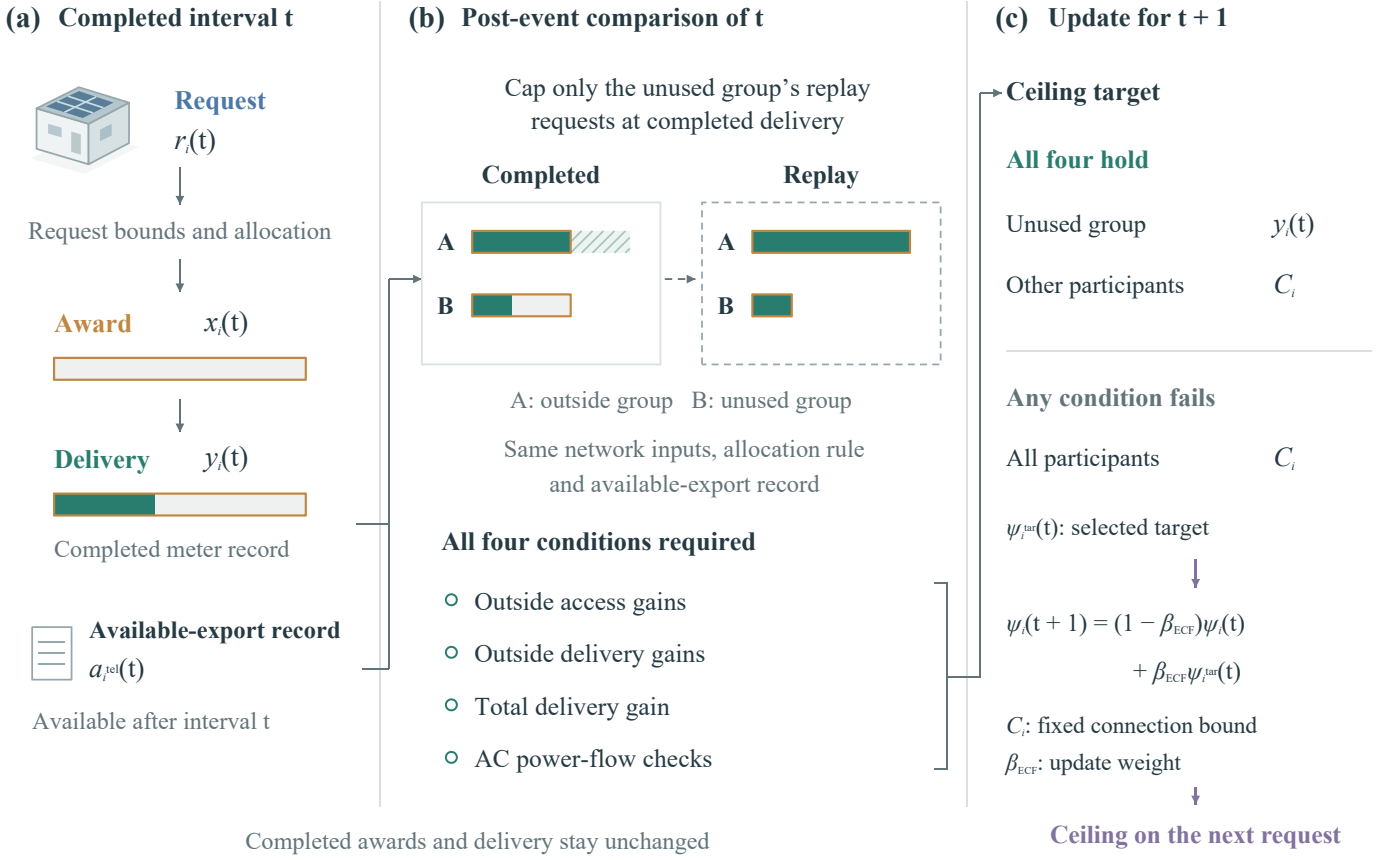


Fig. 2 Using completed delivery to update later request ceilings. (a) Requests pass through connection bounds and current ceilings before the existing allocation rule grants access in interval t . (b) After delivery, only the unused-award group's replay requests are capped at completed delivery. Exogenous network data, rule parameters, and the available-export record stay fixed; awards and power flows are recalculated. The bars follow Fig. 1(c). Qualification requires increased outside access, outside delivery, and total delivery, together with the specified AC checks. Outside gains sum positive changes; the total uses the signed system-wide change. (c) A qualifying replay selects completed delivery as the unused group's target and fixed connection bounds for everyone else. Otherwise, all targets are fixed bounds. The gain β_{ECF} controls the move toward each target. Only ceilings carry forward to $t + 1$; completed awards and delivery remain unchanged.

The Australian public-data simulation gives the same direction of change, with a 1.39-MWh advantage over 80 assessed days. The same daily simulation also tests whether the replay criterion adds value beyond the update formula. Applying that formula after every unused award leaves less authorization unused but delivers 6.42 MWh less than ECF. Thus neither the total request reduction nor the amount of unused authorization alone explains the delivery outcome.

The request-choice analysis first identifies when a larger request benefits its sender at the expense of total delivery. Controlled comparisons then examine which access is preserved by feedback and how this depends on the allocation rule and timing. The public-data results distinguish the effects of request distribution, the four replay conditions, and update weight. Unit gain reverses the benefit, showing why finding recoverable delivery in a completed interval is not enough to ensure a useful subsequent update. Physical checks and equal-total matching are assessed separately from the delivery gains.

Controlled results report mean delivered power in kW. Their request-effect comparisons subtract the forecast-based reference, whereas feedback comparisons subtract the named control from

ECF. The Australian replay reports modeled energy in MWh, with smaller daily and section-level differences in kWh. Positive feedback comparisons favor ECF; all differences are calculated before rounding. Table 3 gives the study map.

3.1 Private delivery gains can accompany a larger loss elsewhere

Larger requests can benefit the requesting participant while reducing total delivery. Across 1,500 controlled parameter–location cases without a shortfall cost, the participant gains 5.2 kW of expected delivery on average while others lose 18.8 kW. Total delivery falls by 13.5 kW. The request model values the participant's own gain but does not include the larger loss elsewhere. Because the same uncertain-output model determines both request choice and expected delivery, this difference arises from the allocation response rather than from unrelated request and output curves.

Fig. 3 shows how this response changes with forecast uncertainty σ and normalized shortfall cost ϑ . The latter is the modeled cost per unit of unused award divided by the value per unit of delivered export. We compare weighted proportional (WP), equal reduction (ER), equal access (EA), maximum export (ME),

Table 3 Study questions, conclusions, and applicable scope

Question	What the experiment establishes	Applicable scope
When does an unused award displace delivery?	Controlled static comparisons (Fig. 3) separate productive high requests from cases in which the changed-request group obtains a higher combined modeled payoff and a larger unused award while total delivery falls.	Designed request comparisons under fixed network conditions.
What does selective feedback add?	Shortfall costs suppress both displacement and useful delivery. At zero shortfall cost and the primary gain, ECF improves mean delivery over mismatch-only feedback and equal-total reduction (Figs. 4–6).	Controlled profiles with exact simulated available-export records.
When can ECF help or harm?	Changing only the time order changes delivery; unit gain can reverse the benefit. Delivery and allocation equality need not move together (Figs. 6–8).	Five rules within a declared synthetic envelope; all reported controlled states pass the specified AC checks.
Does the delivery gain extend to public-data simulations?	The primary replay increases delivery relative to no feedback and equal-total reduction (Fig. 9). Gain, record, and cohort changes show when that advantage persists or reverses (Fig. 10).	Published network and measured profiles; constructed requests and modeled delivery.
What do the replay test and later response each contribute?	Qualification improves delivery over the same update without it, but conservative recovery tests do not always improve subsequent delivery (Sections 3.4.2–3.4.3; SI S10.6, S10.8–S10.9, and S10.15).	Paired network days; continuous memory within observed date blocks.
Who gains, and how do location and PV capacity matter?	Profile placement, PV capacity, and topology change aggregate gains. Individual and section-level gains are not universal (Fig. 11; SI S10.10–S10.14).	Fixed participant locations in placement/capacity tests; separate section, model, and participant outcomes.

and fairness-aware (FA) allocation under two reactive-power settings. Each selected request is paired with the same participant's forecast-based reference. Methods gives the five-outcome available-export model and exact finite-difference calculation.

The aggregate loss does not make every larger request unproductive. We classify a request–access–delivery externality when expected payoff and unused award both rise while total delivery falls. A productive high request instead raises payoff without lowering total delivery beyond the numerical tolerance. For $\vartheta \geq 0$, the externality conditions imply that the requesting participant delivers more and others lose still more; Methods and Section S2 of the *Supporting Information*[†] prove this implication and specify the classification.

Both high-request classes occur in Fig. 3. Among the 1,500 controlled parameter–location cases at $\vartheta = 0$, pooled over all five uncertainty levels, 71.5% meet the externality definition and 11.9% are productive high requests. Reference requests account for 14.7%, and the remaining 1.9% are larger requests that meet neither high-request definition. Section S2 reports the corresponding counts. The coexistence of these classes explains why request size alone is insufficient to identify a delivery loss.

Uncertainty and the normalized shortfall cost shift the relative frequency of these outcomes in opposite directions. With greater forecast uncertainty, a large award can accommodate a higher-output realization. Greater ϑ makes unused awards more costly, so the selected request falls. At $\vartheta = 10$, 10.7% of the 1,500 cases meet the externality definition. Across all cases, the mean total-delivery loss, counting cases without a loss as zero, falls to 0.6 kW. Productive high requests still account for 11.5% of the cases. A shortfall cost can therefore reduce requests that lower total delivery while preserving some productive requests above the forecast.

Panel (d) asks how much delivery is lost elsewhere per extra unit of unused award. For each 0.1 increase in the request multiplier, it divides the first quantity by the second. This ratio also

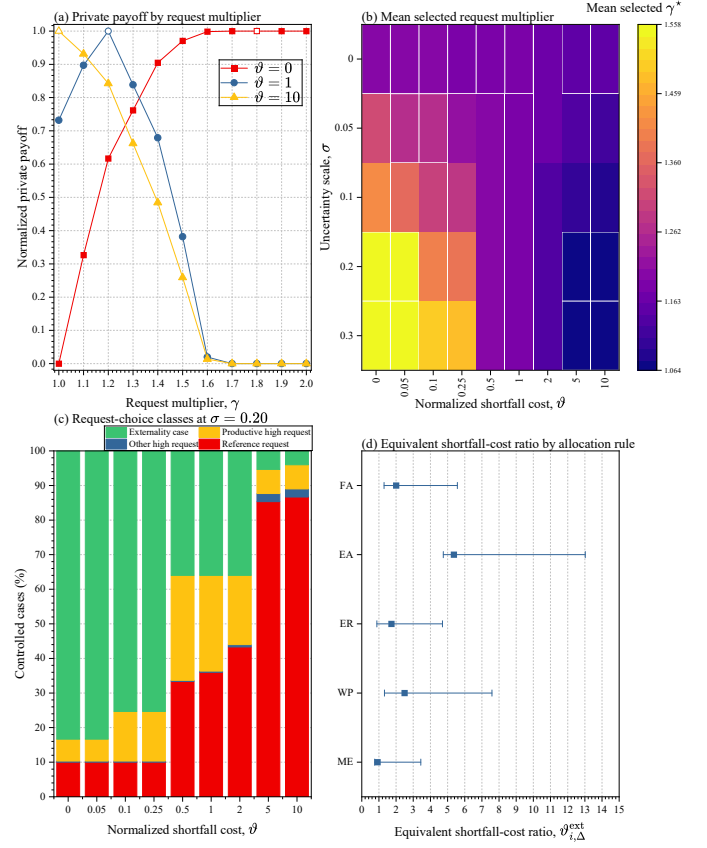


Fig. 3 How a larger request becomes productive or harmful before feedback. (a) Representative private-payoff curves for three normalized shortfall-cost levels. Within each curve, payoff is rescaled as $(U - U_{\min})/(U_{\max} - U_{\min})$; a zero span is plotted as zero. (b) Mean selected request multiplier for each uncertainty and shortfall-cost setting. The mean covers participants, allocation rules, and reactive-power settings. (c) Shares at $\sigma = 0.20$ ($n = 300$): reference requests; productive high requests, which raise payoff without lowering total delivery; request–access–delivery externality cases, which raise payoff and unused awards while lowering total delivery; and other high requests, which meet neither high-request definition. (d) Finite-difference cost ratio $\vartheta_{i,\Delta}^{\text{ext}}$: expected delivery loss elsewhere per unit increase in expected unused award for a 0.1 request-multiplier step. Squares show rule-specific medians, and bars show interquartile ranges. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively.

gives the normalized shortfall cost that would align the participant's payoff change with the total-delivery change. Methods denotes it $\vartheta_{i,\Delta}^{\text{ext}}$ and derives its relation to the local marginal cost. The calculation covers 4,608 steps in which own delivery and unused award rise while total delivery falls. The median ratio is 2.29, with an interquartile range of 1.10–5.72. Rule-specific medians range from 0.92 for ME to 5.37 for EA. This spread shows that one system-wide shortfall cost cannot match the cross-participant delivery effect across all tested electrical locations and operating states. It motivates the post-event feedback comparison in the next section.

3.2 Selective updates preserve delivery that broad reductions remove

At $\beta_{\text{ECF}} = 0.5$ and $\vartheta = 0$, ECF delivers 24.8 kW more on average than the equal-total control. Thus its advantage over uniform reduction cannot be explained by removing more total request. The shortfall-cost and mismatch-only comparisons ask whether request incentives or own delivery provide the same benefit. A representative trajectory first shows the timing needed to interpret these comparisons: feedback changes a later allocation, not the completed interval used for the replay. All dynamic controls start from the same raw requests and allocation in the first interval. The changed-request group is fixed for each trajectory, whereas the unused-award group is identified anew after each interval and determines whose requests ECF caps in the replay. Fig. 2 shows the timing, and Methods and Section S3 of the *Supporting Information*[†] define the update.

3.2.1 Completed delivery affects later access

Feedback acts with a delay: it cannot change the interval whose shortfall prompted the update. This is visible in the representative trajectory in Fig. 4, where the controls start from the same allocation and diverge only after a completed record becomes available. The three 16-interval profiles use standardized forecast deviations $-2, -1, 0, 1, 2$ with counts 1, 4, 6, 4, 1. They differ only in order: clustered groups the low states, intermittent separates them, and shock-recovery places a low-output episode between initial normal output and recovery. Participant offsets retain these counts while preventing perfectly synchronized output. Section S3 gives the exact sequences.

Fig. 4 follows one trajectory in which five electrically nearby participants change their requests and the available-export states follow a shock-recovery ordering. It uses WP allocation, zero participant reactive power (Q_0), and $\vartheta = 0$. An unused award first supplies the candidate for replay; a feasible increase in delivery then supports a later ceiling adjustment. The distinction between panel (c)'s completed-state recovery and panel (d)'s later delivery is important: the former justifies attempting an update, whereas the latter measures its outcome after available export changes.

3.2.2 A shortfall cost reduces displacement but also useful delivery

An anticipated shortfall cost reduces the incentive to request unused access, but also gives up some delivery during high-output intervals (Fig. 5). We measure that trade-off against forecast-based reference requests. The externality-interval share counts the economic classification defined above; ECF activation counts a different event, the completed-state replay meeting its four conditions.

With the shortfall cost alone, raising ϑ from 0 to 1 lowers the mean request multiplier from 1.51 to 1.14. The externality-interval share falls from 50.1% to 15.6%, and delivery loss outside the changed-request group falls from 37.6 to 9.4 kW. The cost also removes 45.6% of that group's delivery gain, reducing it from 15.9 to 8.7 kW. At $\vartheta = 10$, no interval meets the externality definition, but the group's gain is only 0.07 kW. A strong shortfall cost therefore reduces requests associated with delivery loss else-

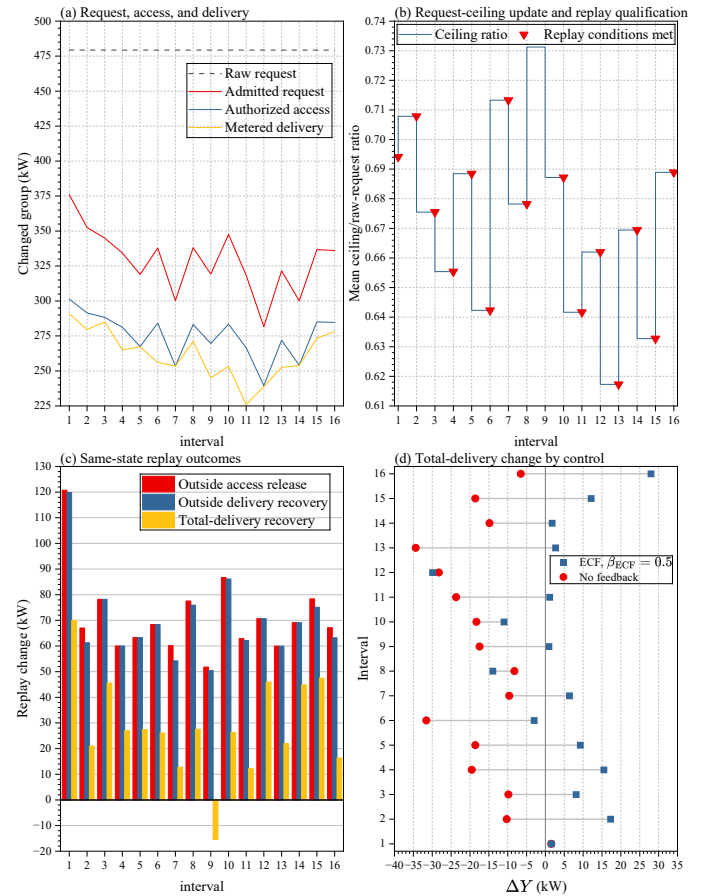


Fig. 4 One 16-interval ECF trace for the local-five, shock-recovery, WP-Q0 case at $\vartheta = 0$. (a) Raw request, admitted request, awarded access, and metered delivery for the changed-request group. (b) Mean ceiling-to-raw-request ratio after each update. Red triangles mark intervals that meet the replay conditions. (c) Access released and delivery recovered outside the current unused-award group, with the signed change in total delivery. (d) Total-delivery change under ECF and no feedback. The controls coincide in interval 1 because no earlier meter record exists. The local-five case changes five electrically nearby participants; shock-recovery is one 16-interval ordering of the same available-export-node frequencies. WP denotes weighted proportional allocation, and Q_0 sets participant reactive power to zero.

where, but it also removes useful access when available export is unexpectedly high.

At $\vartheta = 0$, the shortfall-cost benchmark coincides with no feedback. Using that common baseline, mismatch-only feedback increases mean total delivery by 15.2 kW, while ECF increases it by 24.9 kW. ECF's additional advantage over mismatch-only feedback is therefore 9.7 kW. The three controls receive the same raw requests. Relative to forecast-based reference requests, the no-feedback loss of 21.7 kW becomes a gain of 3.2 kW under ECF. ECF retains 79.4% of the changed-request group's no-feedback delivery gain and reduces the delivery loss outside that group to 9.4 kW. The externality-interval share falls by 29.0 percentage points. Section S3 reports the group decomposition and the control comparisons with their respective baselines.

A gain in average delivery can coexist with more intervals classified as request-access-delivery externalities because a count does not reflect the magnitude of delivery changes. At $\vartheta = 1$,

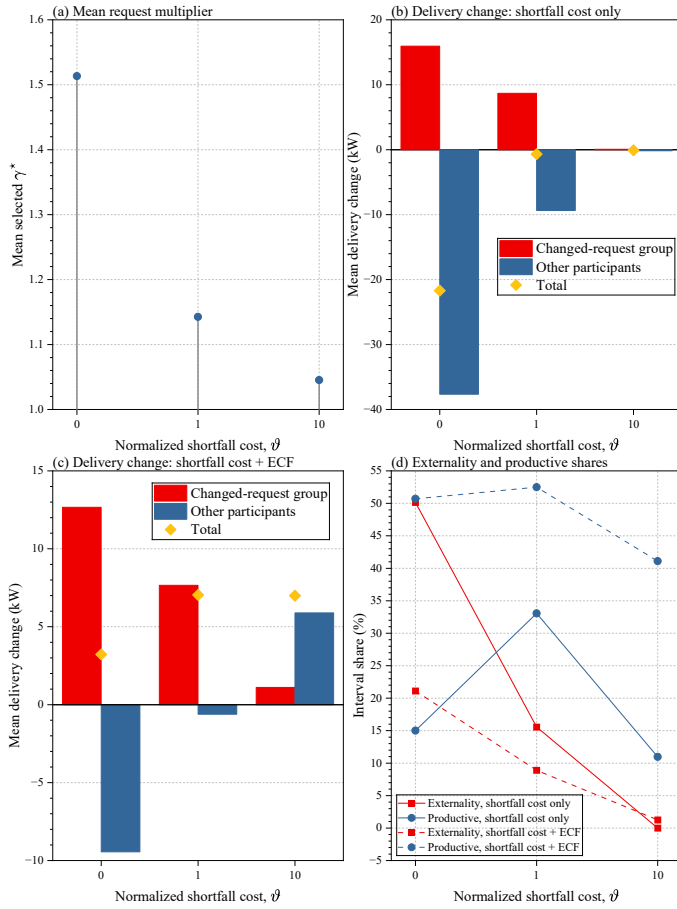


Fig. 5 How a modeled pre-allocation shortfall cost and post-delivery ECF affect requests and delivery. Results cover all profiles, spatial cases, allocation rules, and reactive-power settings. (a) Mean selected request multiplier. (b) Delivery changes with the shortfall cost alone. The bars separate the changed-request group and other participants; diamonds show their total. (c) The same changes with the shortfall cost and ECF at $\beta_{ECF} = 0.5$. Positive blue bars denote recovery outside the changed-request group. (d) Externality and productive-high-request shares with and without ECF.

ECF increases total delivery by 7.7 kW relative to the shortfall-cost benchmark and lowers the externality-interval share by 6.7 percentage points. At $\vartheta = 10$, the benchmark already has no externality interval. Adding ECF raises the share to 1.3%, although mean delivery still increases relative to that benchmark. Feedback can therefore improve average delivery even when that economic classification becomes more frequent.

3.2.3 Where requests are reduced determines recovered delivery

ECF and the equal-total control remove the same total request in every interval but reduce different participants' requests. The comparison holds the allocation rule, exogenous network conditions, and available output fixed. It measures the delivery effect of the request distribution produced by ECF relative to proportional reduction. Both electrical location and the output available at each connection contribute to that effect. Mismatch-only feedback provides a separate comparison with an update based on own delivery, without testing recovery elsewhere. It changes both the information used and the update law, whereas the equal-total

control fixes the amount removed. The two comparisons therefore assess the resulting updates from complementary directions.

At $\vartheta = 0$ and $\beta_{ECF} = 0.5$, ECF delivers 9.7 kW more than mismatch-only feedback and 24.8 kW more than the equal-total control. ECF delivers more in 73.3% and 95.6% of the matched trajectories ($n = 90$ for each comparison), respectively. At $\vartheta = 1$, the mean advantages are 6.8 and 9.0 kW. At $\vartheta = 10$, they are 6.8 and 8.2 kW. The advantage over both controls thus persists across the three tested shortfall-cost settings.

A favorable replay does not guarantee a favorable later update. The feedback gain controls how much weight the latest interval receives. Delivery changes nonmonotonically with that weight. At $\vartheta = 0$, the mean change relative to forecast-based requests is -2.8 kW for $\beta_{ECF} = 0.25$. It rises to 3.2 kW at 0.5 but falls to -23.4 kW at 1 . The same reversal occurs at $\vartheta = 1$ and 10 . The replay evaluates a completed output realization, whereas the resulting ceiling applies to a different interval. At unit gain, the previous ceiling is discarded in favor of that completed interval's target. The negative response establishes the importance of this temporal distinction; it does not by itself identify a single cause of the reversal. We use 0.5 as the primary intermediate update weight, which gives the largest mean delivery among the three tested gains in all three shortfall-cost settings.

The controlled study uses the exact available-export record in Assumption 1; here, that record equals the simulated available export. Of 12,960 replay attempts, 12,910 meet all four conditions. Among these intervals, median total-delivery recovery is 37.8 kW. The interquartile range is 25.6–57.5 kW, and 99.5% of recoveries exceed 5 kW. At $\beta_{ECF} = 0.5$, 20 of 4,320 attempts do not meet the conditions because total delivery does not rise. An unused award therefore initiates a replay but does not by itself lower a later request ceiling. These controlled trajectories combine larger requests with constrained export access, so the replay usually finds delivery that could be recovered elsewhere. The Australian replay has a lower activation rate of 12.6%. The two rates describe how often the four conditions hold in their respective experiments.

Fig. 6 reports the gain sweep, paired control comparisons, and replay-recovery distribution behind these main comparisons.

3.3 Timing and allocation change the size of the delivery gain

At $\vartheta = 0$ and $\beta_{ECF} = 0.5$, the mean ECF delivery advantage is positive at every factor level reported in Fig. 7. This holds for every allocation rule, temporal profile, spatial case, and reactive-power setting. Against no feedback, the rule-specific gain ranges from 8.9 kW for FA to 67.5 kW for EA. Against the equal-total control, it ranges from 9.7 to 50.8 kW. The positive direction is therefore not confined to one allocation rule.

The three temporal profiles contain the same available-export values and frequencies. Only their order changes. The mean delivery advantages over no feedback are 35.3 kW for the clustered profile, 12.2 kW for the intermittent profile, and 27.3 kW for shock-recovery. Against the equal-total control, the corresponding advantages are 34.4, 12.6, and 27.4 kW. Because each profile preserves the available-export values and their frequencies, these

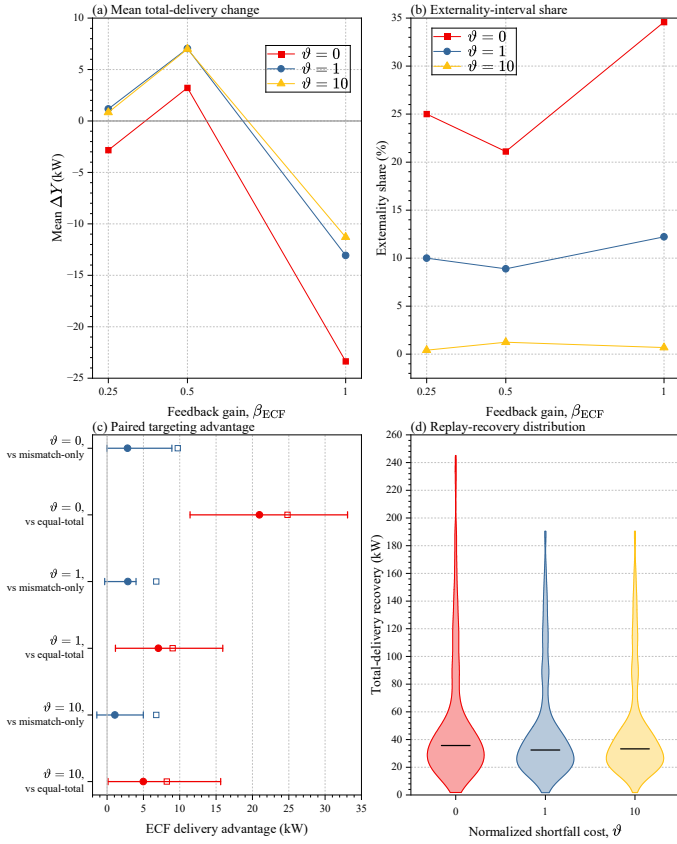


Fig. 6 Feedback-gain response, paired controls, and replay recovery across all controlled dynamic cases. (a) Mean signed change in total delivery and (b) externality-interval share for the three tested feedback gains. The series denote $\vartheta = 0, 1, 10$. (c) ECF delivery advantage in matched trajectories. Rows labelled “vs mismatch-only” compare ECF with mismatch-only feedback. Rows labelled “vs equal-total” compare ECF with the equal-total control. Panel (c) uses the primary gain, $\beta_{ECF} = 0.5$. Filled circles show medians, open squares show means, and whiskers show interquartile ranges. The gray line marks zero. (d) Distribution of total delivery recovered when the replay meets all four conditions. Violin width shows empirical density, and the black horizontal segment shows the median.

differences isolate temporal ordering rather than a change in the uncertainty distribution. A completed low-output interval has a different value for later access when similar states are clustered than when they are separated by other output states.

The mean delivery advantage over no feedback is 11.2 kW when one participant changes its request. It is 35.0 kW for five electrically local participants and 28.6 kW for five dispersed participants. Every participant still selects its own request multiplier. The local-dispersed comparison holds group size fixed but changes participant identities, electrical locations, and participant-specific multipliers. The difference therefore reflects the selected participants as well as their electrical placement. At the reported precision, zero reactive power (Q0) and 0.95-power-factor absorption (Q95) both give a 24.9-kW mean advantage over no feedback. Both also give a 24.8-kW mean advantage over the equal-total control. The two comparator-specific results are therefore unchanged across the two tested reactive-power settings.

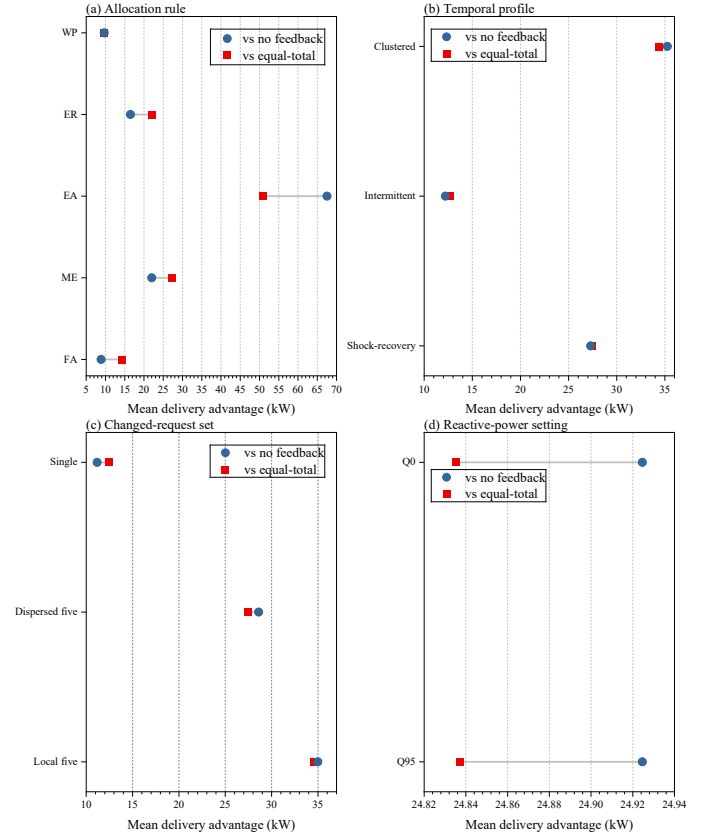


Fig. 7 Mean ECF delivery advantage at $\vartheta = 0$ and $\beta_{ECF} = 0.5$. Results are grouped by (a) allocation rule, (b) temporal profile, (c) changed-request set, and (d) reactive-power setting. Blue circles compare ECF with no feedback. Red squares compare ECF with the equal-total control. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively. Q0 sets participant reactive power to zero, whereas Q95 uses 0.95-power-factor absorption.

3.3.1 Feasible delivery gains do not imply equal access

The delivery comparisons remain within the declared physical envelope. Both sides of all 25,920 matched allocation comparisons pass nonlinear AC checks for convergence, active- and reactive-power balance, voltage, and two-ended branch apparent power. Since delivery can be below the award, we separately check the resulting connection injections. All 22,520 distinct observed- and replay-delivery states pass the same checks, with maximum branch utilization of 90.91% and no excess above the fixed synthetic ratings. Section S3.5 of the *Supporting Information*[†] gives the state counts and voltage extrema. Fig. 8(a) shows the voltage margin: the minimum bus-voltage magnitude minus the 0.95-per-unit (pu) lower limit. Physical feasibility thus holds for both authorized and delivered export, including cases where feedback reduces delivery.

More delivery does not automatically mean more equal access. Fig. 8 therefore reports both outcomes. The Jain index measures capacity-normalized allocation equality, not broader social fairness. At $\vartheta = 0$, ECF yields a higher mean Jain-index change than no feedback under all five rules, but most changes remain negative relative to the forecast-based reference allocation. For EA, the change moves from -2.4 percentage points without feedback to

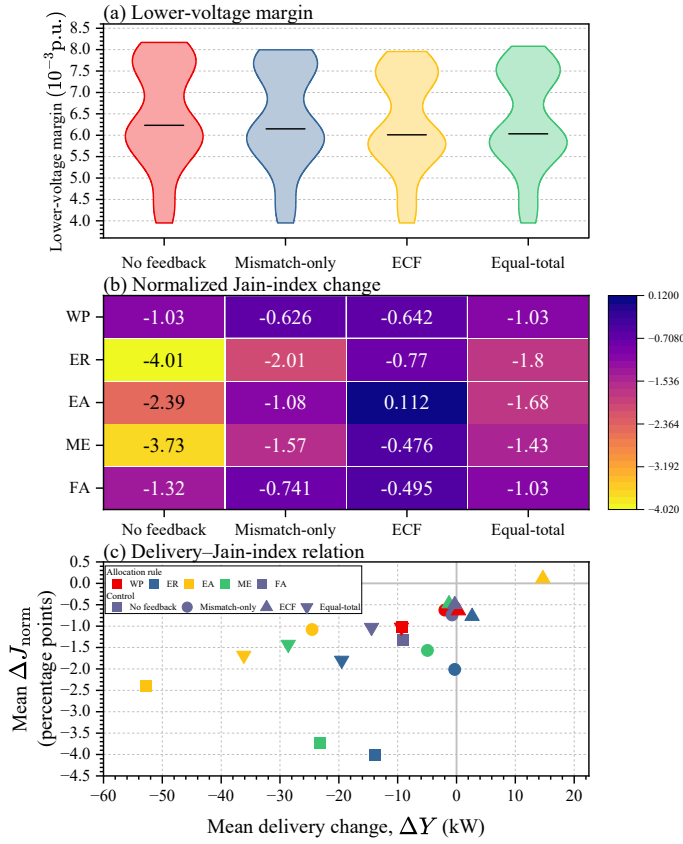


Fig. 8 Network-limit and allocation-equality outcomes for the AC-validated controlled states. (a) Lower-voltage margin at $v = 0$, where the margin is the minimum bus-voltage magnitude minus the 0.95-pu lower limit. Violin width shows empirical density, and the black segment shows the median. (b) Mean change in the capacity-normalized Jain index, in percentage points, relative to the forecast-based reference allocation, by rule and control. Negative values indicate lower allocation equality than that reference. (c) Mean total-delivery change plotted against the corresponding Jain-index change. Colors identify allocation rules, and marker shapes identify no feedback, mismatch-only feedback, ECF, and the equal-total control. ECF and equal-total use $\beta_{ECF} = 0.5$.

0.1 percentage points with ECF. The ECF values for WP, ER, ME, and FA range approximately from -0.8 to -0.5 percentage points: ECF reduces the decline rather than producing an absolute increase under these four rules. ECF also exceeds the equal-total control on this mean metric for every rule. Panel (c) shows that mean delivery and allocation-equality outcomes vary across rule-control combinations. Delivery and allocation equality can therefore move in different directions. The *Supporting Information*[†] reports the corresponding unused-award, allocation-movement, and total-authorization results.

Thus physical feasibility, delivered electricity, and allocation equality answer different questions. The observed gain in delivery is not a claim that ECF maximizes every sharing objective. The *Supporting Information*[†] reports secondary allocation-formulation analyses under the same operating limits.

3.4 Public-data replay gains delivery through a different allocation of access

3.4.1 Better use of access increases delivery, not just utilization

ECF also outperforms equal-total reduction when requests and available output follow public household profiles. Across 80 assessed network-days, it delivers 239.36 MWh of modeled export, 1.39 MWh (0.59%) more than each of no feedback and the equal-total control (Fig. 9). The simulation combines a published Australian MV/LV network with measured PV and load chronology. Available export and requests are constructed from these profiles, rather than taken from linked operating records. Allocation uses a common proportional scale, with daily ceiling initialization in the primary comparison. Methods specifies the construction; the following comparisons examine what changes the delivery gain.

More delivery here accompanies more authorization, rather than a smaller total award. Relative to no feedback, authorization increases by 1.206 MWh and unused authorization falls by 0.186 MWh. Together these changes account for the 1.392-MWh delivery gain in (40). Selective request reduction therefore improves delivery without requiring a reduction in total authorization. Section S10 gives the complete energy balance, including the almost identical contrast with the equal-total control.

Increased authorization is the larger term in this energy balance. Equal admitted-request totals do not fix the total award: different request distributions can admit different proportional allocation factors. Conversely, uniformly scaling requests preserves their relative sizes. If the network constraint still binds, the allocation factor can compensate for that scaling and leave awards unchanged. The near-equality of no-feedback and equal-total delivery is consistent with this property, derived in Section S4. Selective updates change relative requests instead. The energy balance describes their resulting delivery gain without assigning independent causal effects to increased authorization and reduced unused awards.

The gain is uneven over time. ECF delivers more on 44 days, less on 32, and the same amount on four against each control. Its mean daily gain is 17.4 kWh, with dependence-adjusted 95% intervals of 5.1–29.7 kWh against no feedback and 4.9–29.9 kWh against equal-total reduction. Only 483 of 3,840 half-hours (12.6%) qualify for tightening. This frequency describes how often the model finds a recoverable opportunity, not detection accuracy or a guarantee of improvement on every day.

The connection-point gain also appears at the source terminal. Net source exchange increases by 1.377 MWh against each control, retaining about 98.9% of the respective delivery gains. The feeder remains a net importer, so this change means reduced upstream import. It includes the small increase in circuit losses and changes in voltage-dependent passive-element exchange; Section S10 separates those quantities. The delivery gain is therefore visible at the feeder boundary, not only in the connection-level allocation totals.

Authorization utilization, E_Y/E_X , is 72.6% under ECF and 72.4% under each of no feedback and the equal-total control. Here E_Y and E_X are delivered and authorized export energy over

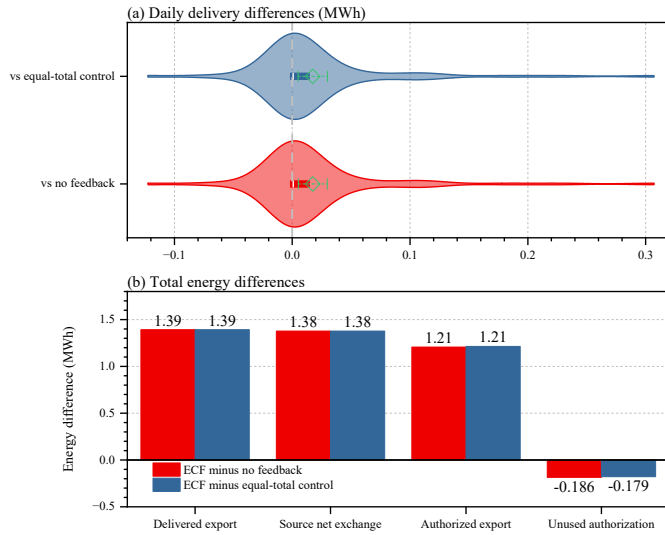


Fig. 9 Modeled delivery and authorization changes in the Australian public-data replay. Available export is constructed from PV and load profiles. (a) Paired daily delivery differences for ECF against no feedback and the equal-total control. Violin width shows empirical density. Dark boxes span the 25th–75th percentiles, and white ticks show medians. Green diamonds show means; capped green lines show 95% intervals adjusted for short-range dependence between neighboring dates. (b) Aggregate 80-day differences in delivered export, source-terminal net exchange, authorization, and unused authorization. Each difference is ECF minus the indicated control. Delivery equals authorization minus its unused portion. Here the delivery gain accompanies increased authorization and reduced unused authorization. Positive source-terminal differences indicate reduced net upstream import over the assessed period.

the assessed days. Previous-day persistence gives a useful contrast: it omits the forecast increment and uses 89.0% of its authorization, yet delivers 50.33 MWh less than ECF. A high utilization ratio alone can therefore conceal delivery lost through an overly restrictive request. This contrast makes utilization insufficient as an operating objective: withdrawing access can improve the ratio even while reducing delivered energy.

All primary control trajectories satisfy the specified allocation and delivery checks. Network limits nevertheless reduce the admitted-request scale in 44.4% of ECF half-hours. The comparison therefore concerns how constrained access is used, rather than a benefit obtained by allowing infeasible delivery. Fig. 9 reports the daily differences and aggregate energy changes.

3.4.2 Replay checks and update weight have distinct effects on delivery

A same-update comparison isolates the value of checking recoverable delivery. Removing the replay test while keeping the update formula and gain unchanged reduces modeled delivery from 239.36 to 232.93 MWh. ECF therefore delivers 6.42 MWh (2.76%) more, with higher delivery on 75 of 80 days. Its mean daily advantage is 80.3 kWh, with a dependence-adjusted 95% interval of 69.3–91.2 kWh.

The control that skips these checks leaves 18.11 MWh less authorization unused, but also grants 24.54 MWh less authorization. Reducing unused awards more aggressively therefore sacrifices useful delivery. This comparison attributes the difference

to applying the replay criterion within the same recursive update, rather than to a different smoothing formula. Both controls pass the specified physical checks. Section S10.6 reports their energy totals, participant-level differences, and the additional computation needed for replay.

Qualification determines whether to tighten the unused group’s ceilings; gain determines how strongly they respond. At $\beta_{\text{ECF}} = 0.25$, ECF gains 1.34 MWh over each of no feedback and the equal-total control, slightly less than at the primary gain of 0.5. At $\beta_{\text{ECF}} = 1$, it loses 3.04 MWh relative to no feedback and 0.54 MWh relative to the equal-total control. The first mean-difference interval is below zero; the second includes zero. Full replacement of the ceiling by the latest target therefore offers no delivery advantage here. Relative to no feedback, unit gain reduces total authorization by 14.55 MWh and unused authorization by 11.51 MWh. Their difference explains the delivery loss: some of the authorization removed would have been used.

After a qualifying replay, unit gain sets each unused-award participant’s next ceiling to its completed delivery. The largest net losses relative to gain 0.5 occur at connections whose available output then rises while the new ceiling still limits their request (Section S10 of the *Supporting Information*[†]). Together with the controlled gain sweep, this separates two requirements: the completed-state comparison must be valid, and the later update must suit the evolution of available export. Satisfying the first does not ensure the second.

The gain reversal persists when ceilings carry across midnight. In ten consecutive-date blocks totaling 80 days, the longest spanning 35 days, total delivery at each of the three gains differs from daily initialization by less than 0.1 kWh. The primary advantage and the unit-gain loss are therefore not explained by the daily reset. Section S10.8 relates ceiling recovery to the update law and reports every block, including three in which gain 0.5 delivers less than no feedback.

ECF also retains its aggregate delivery advantage when participants adjust requests from their own past delivery. Each participant chooses among smaller and larger requests, with preferences and ceilings carried across consecutive days. Across the 80 dates and three random seeds, ECF delivers 1.44 MWh more than no feedback and 4.53 MWh more than the same update without qualification; both controls develop their own request histories. The equal-total shadow, which follows ECF’s history rather than learning independently, delivers 1.50 MWh less than ECF. These totals average the seeds within each date. Compared with keeping request-choice probabilities fixed, preference updates increase the advantage of retaining the replay checks, but do not establish a larger advantage over no feedback. Section S10.15 reports the request model, paired uncertainty, and daily variation. Thus ECF’s mean benefit survives experience-based request adjustment, without requiring requests to remain fixed.

3.4.3 Record quality affects qualification and later delivery differently

The mean delivery gain tolerates several record perturbations, but this does not establish that every replay remains correct. The four perturbations scale the post-event record by $\pm 10\%$, delay

it by one interval, or make a deterministic 10% pattern of participant records unavailable and substitute modeled delivery for those entries. Mean gains range from 16.4 to 18.0 kWh per network day against no feedback and from 16.3 to 18.0 kWh against the equal-total control. Every dependence-adjusted 95% interval remains above zero. The delivery comparison is therefore less sensitive to these record changes than to the gain change. A separate lower-bound calculation shows that individual completed-interval comparisons have widely varying margins to record error (Section S10 of the *Supporting Information*[†]). Thus robustness of the mean gain does not imply equal robustness of each replay decision.

Using a delivery lower bound protects the recovery test, but does not always increase later delivery. A separate comparison runs both record treatments at error radii of 1%, 5%, and 10% of registered capacity, under common overestimation and underestimation. Point-record control tightens after 11, 28, and 33 intervals in the three overestimation cases even though the unperturbed model does not meet the recovery criteria. Lower-bound control has no such interval in any of the six cases.

Underestimation creates the opposite tradeoff: at the largest radius, lower-bound control misses 258 of 496 feasible recovery opportunities and delivers 269 kWh less than point-record control. The mean daily difference is -3.36 kWh, with a 95% interval of -6.39 to -0.34 kWh. Across all six error cases, lower-bound control still delivers 0.95–1.46 MWh more than no feedback, with positive mean-difference intervals. All resulting allocation, delivery, and reference replay states pass the specified physical checks. Section S10.9 of the *Supporting Information*[†] gives the complete comparisons. A reliable basis for an update and a beneficial future response are thus separate requirements, even when the record error is bounded.

Fig. 10 collects the gain and record responses, together with the complete-customer cohort. The latter broadens the calendar to 361 days at the same load multiplier of 0.30. ECF gains 10.96 MWh over each of no feedback and equal-total reduction, or 30.4 kWh per network day; both mean-difference intervals remain above zero and all specified physical checks pass. This result extends the positive comparison to a different customer cohort and fuller calendar. It does not isolate record coverage, because customer composition changes as well (Section S10).

3.4.4 Spatial placement and available generation shape the benefit

The delivery advantage persists when the same household profiles occupy different service connections. Across five alternative mappings, ECF gains about 0.9–5.1 MWh over either no feedback or the equal-total control during the same 80 days. All ten pointwise 95% intervals for mean daily differences are above zero. The positive aggregate comparison therefore extends beyond the primary profile assignment, although its size varies substantially with placement.

The smallest aggregate gain also illustrates why unused authorization alone cannot measure success. In that mapping, ECF grants more authorization than no feedback and leaves 0.35 MWh more unused, yet delivers 0.86 MWh more. It gains delivery on

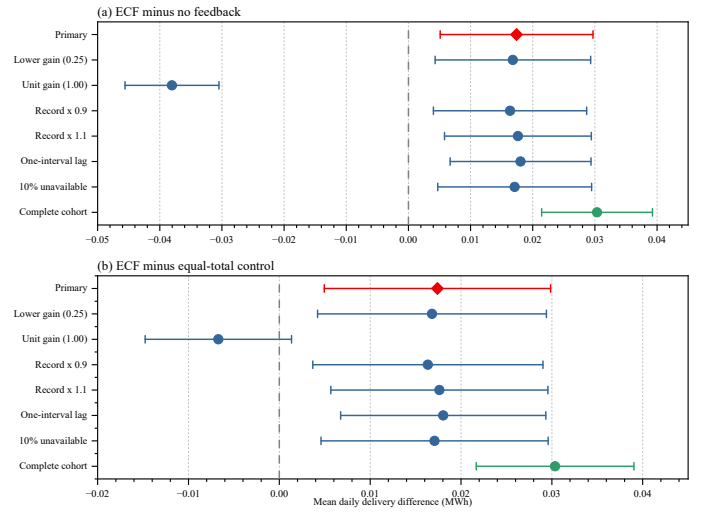


Fig. 10 Sensitivity of the Australian replay to feedback gain, post-event record, and customer cohort. Markers show paired mean delivery differences per network day. Bars show 95% intervals adjusted for short-range dependence between neighboring dates. (a) ECF minus no feedback. (b) ECF minus the equal-total control. The red diamond denotes the primary unperturbed-proxy result, and blue circles denote gain or record changes. Green markers denote the complete-customer cohort, which uses the same load multiplier but changes customer composition and calendar coverage. Every transformed available-export record is floored at modeled delivery. All settings pass the specified allocation and delivery checks; the zero line separates positive from negative mean delivery differences.

24 days and loses on 43, with no change on 13; larger positive days outweigh the more frequent losses. Section S10.10 of the *Supporting Information*[†] gives each mapping's energy totals, daily differences, and qualifying intervals. All allocation, delivery, and replay states pass the specified physical checks.

The advantage also persists across PV capacities. At five levels spanning half to one-and-a-half times the primary capacity, ECF gains 1.39–1.88 MWh over either no feedback or the equal-total control during the same 80 dates. Every pointwise mean-difference interval is above zero. The gain does not increase monotonically with installed capacity, even though network limits restrict requests more often. Relative to no feedback, delivery gained per installed MW falls from 0.81 to 0.27 MWh across the smallest and largest capacities. More PV therefore does not imply a proportionally larger benefit from this feedback.

Capacity also changes the balance between authorization and its use. At half the primary capacity, ECF grants 2.26 MWh more authorization and leaves 0.60 MWh more unused, yet delivers 1.66 MWh more than no feedback. At the primary capacity, unused authorization instead decreases. Both responses improve delivery, but only one reduces unused authorization. Section S10.11 of the *Supporting Information*[†] gives all four controls, daily comparisons, and their satisfactory physical checks. Together, the placement and capacity comparisons show why delivered energy is the relevant outcome: unused authorization can rise or fall while delivery improves.

Changing the network section gives a related spatial comparison. ECF delivers more than the equal-total control in all 13 published LV sections. The 80-day differences range from 71.2 to

1,311.6 kWh. Relative to no feedback, 12 sections gain delivery; LV26 loses 3.7 kWh. All sections contain qualifying replay intervals. Selecting where requests are reduced therefore improves on uniform reduction throughout this panel, but it does not always improve on leaving requests unchanged.

All section trajectories satisfy the specified allocation and delivery checks. Each section uses its own published model with the primary chronology, profile mapping, load multiplier, authorization voltage ceiling, and $\beta_{ECF} = 0.5$. These 13 sections belong to one parent network collection. Their separate-model gains are not added to estimate the coupled feeder's gain. The *Supporting Information*[†] reports every section's delivery difference, activation count, and physical-check status.

The representative-topology panel also contains null responses. Six of the nine representative LV models contain qualifying replay intervals and show positive aggregate delivery differences against both controls. The models—G, J, L, M, N, O, P, Q, and R—come from the National Low-Voltage Feeder Taxonomy Study⁴². All four control trajectories pass the specified physical checks in every model. J and Q have the largest gains: 543.0 and 960.5 kWh against no feedback, and 611.7 and 982.1 kWh against the equal-total control.

G, M, P, and R also have positive differences, although M qualifies only twice and gains just 0.1 kWh over no feedback. L, N, and O have no qualifying intervals and remain unchanged. The common feedback setting therefore produces substantially different responses across the published structures. These results extend the positive comparison beyond one topology while identifying models where no update is warranted by the replay. Methods gives the electrical inclusion criteria. Fig. 11 and the *Supporting Information*[†] report each model separately.

Spatially uneven gains also appear within the primary feeder. Of its 418 participants, 330 deliver more over the 80 days than without feedback and 88 deliver less; the largest individual loss is 7.3 kWh. Section S10.14 reports the cumulative and daily distributions. A positive feeder total therefore does not imply that every connection benefits, just as a favorable mean across days does not imply improvement on every day.

3.5 Implications for the use of constrained export access

The findings suggest that export-access arrangements should be evaluated by the electricity delivered as well as the permission allocated. Persistence uses a larger fraction of its award but delivers substantially less electricity. The same update without replay qualification removes more unused authorization than ECF, yet also delivers less. Neither minimizing unused awards nor maximizing authorization utilization is therefore a sufficient operating objective. A useful adjustment must preserve access that can be used when output recovers.

Use completed records to complement allocation and settlement. Network-aware allocation already accounts for electrical location. The equal-total comparison shows why later request adjustments also need to consider where available electricity remains: an identical aggregate request reduction can produce different awards and delivery. Completed records provide a way

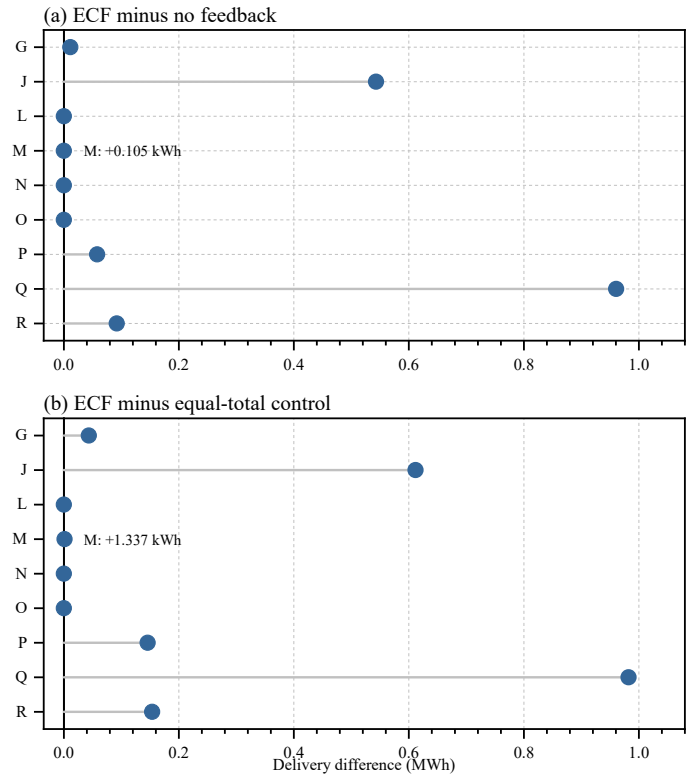


Fig. 11 Delivery differences across nine published representative low-voltage models. Each marker is the 80-day ECF-minus-control total for one model; horizontal stems connect zero to that total. (a) ECF minus no feedback. (b) ECF minus the equal-total control. All models pass the specified voltage and transformer checks across the four controls. Six models have positive totals; L, N, and O have no qualifying interval and zero difference. Qualifying-interval counts are reported separately in the SI. Results are not pooled across models.

to test that opportunity while retaining the chosen sharing rule. This is a distinct role from shortfall settlement, which changes the incentive to request access. The state-dependent relationship in Fig. 3(d) explains why one shortfall cost cannot represent every loss of delivery elsewhere.

Flexible export arrangements provide a relevant setting for this use of operating information^{1–3}. The proposed addition is a basis for updating requests, not a replacement for network limits or a new financial settlement. Source-terminal accounting shows that the resulting connection-level gain can also reduce upstream import after network losses and passive-element exchange are included. This connects the allocation decision to feeder energy exchange.

Separate the reason to tighten from the speed of response. A favorable replay identifies a missed opportunity in the completed interval. It does not determine how much access the same participants will need next. The gain reversals, recovery trajectories, and cross-day comparison show why these are different questions. A large response to low completed delivery can restrict export when available output rises. Gain selection should therefore use subsequent delivered energy and ceiling recovery, rather than the frequency of qualifying replays. Evaluation against both no feedback and equal-total reduction is important: the LV26 comparison improves on uniform reduction but not on leaving requests

unchanged.

Track who gains and who loses. The spatial comparisons show variation in both aggregate benefit and its distribution. The primary feeder gains delivery overall, but some participants lose cumulative delivery. Operational assessment should therefore retain participant-level delivery and the duration of restrictive ceilings alongside the feeder total. Replay qualifies the unused-award group collectively; it does not identify each member's causal contribution. Its output supports an operational update, not individual blame or a charge for displaced energy.

Match the information requirement to the intended use.

A pilot needs linked requests, awards, metered delivery, and available-export records, with connection identities and the applicable network model. It need not reveal participants' private forecast distributions. However, available export must remain valid for the alternative allocation; responses of storage, demand, or local controllers must be modeled when access changes them. Independently validated error bounds can make the recovery comparison conservative, while AC feasibility is checked separately. When increased replay delivery cannot be established, ECF lets ceilings recover toward registered capacity. The bounded-record results also show the cost of caution: rejecting more doubtful updates can forgo later delivery gains. Record validation, communication, and replay computation must therefore be assessed against the additional energy delivered, rather than inferred from the activation rate.

Scope and next evidence. The controlled study connects request incentives to delivery under five rules within a declared synthetic medium-voltage envelope. The public simulations add measured household chronology, customer-voltage limits, and native transformer ratings, but construct the profile mapping, requests, and available-export record. Spatial and capacity comparisons vary these conditions without estimating the prevalence of opportunities across feeders. The primary calendar covers 80 seasonally incomplete days; continuous memory is tested within observed date blocks of up to 35 days. Experience-based request adjustment is included, whereas anticipation of future ceilings and strategic manipulation of availability records are not. Linked field records and longer operating histories would test those behaviors and the cost of implementation. Annual energy, market value, household welfare, generation displacement, and emissions require their own calendar coverage and models. The SI reports the physical-check coverage separately from these inference limits: a feasible trajectory can still have an unfavorable delivery outcome.

4 Methods

4.1 Network allocation and information timing

The model separates three stages: an uncertain request, a network-constrained award, and delivery limited by realized available export. Feedback carries information from completed delivery into the next request stage. Let $\mathcal{G} = (\mathcal{B}, \mathcal{L})$ denote the distribution-network graph, and let $\mathcal{N}$ denote the set of participating connections. The map $v(i) \in \mathcal{B}$ assigns participant i to a bus.

In interval $t \in \mathcal{T}$, participant i has a fixed connection bound $C_i > 0$. This bound caps the request admitted to allocation and the resulting award; the raw request may exceed it. The participant observes a private signal $s_i(t)$ about its available active-power export $a_i(t) \geq 0$. Its conditional cumulative distribution function (CDF) is $F_i(\cdot | s_i(t))$. The participant submits request $r_i(t) \geq 0$ before availability is realized. The coordinator applies allocation rule $\mathcal{R}$ under network model $\mathcal{M}$. It grants export access $x_i^{\mathcal{R}, \mathcal{M}}(t)$. Metered delivery is then $y_i(t) = \min\{x_i^{\mathcal{R}, \mathcal{M}}(t), a_i(t)\}$. We suppress the rule and model superscripts when they are clear. The request is based on this available-export model, and delivery is limited by its later realization.

Before allocation, the coordinator does not observe $s_i(t)$, F_i , or $a_i(t)$. Availability is realized during the interval. After completion, the coordinator observes $y_i(t)$ and receives the specified available-export record $a_i^{\text{tel}}(t)$. Here, the superscript tel denotes the post-event record interface, whether the record is measured or constructed from specified data. ECF also uses the admitted-request record, completed allocation, unchanged allocation rule and network model, and bound C_i . The private signal and available-export CDF remain with the participant.

Assumption 1 (Post-event available export in the experiments). *For each completed controlled interval, the data-generating process records connection-point available export after allocation and delivery. The controlled study sets $a_i^{\text{tel}}(t) = a_i(t)$ for every participant. This record is unavailable when $\mathbf{x}(t)$ is selected, so it cannot alter that allocation. The Australian replay instead constructs a profile-derived available-export proxy from scaled gross-PV and household-consumption profiles. That proxy generates modeled delivery. The primary replay also uses its unperturbed value as $a_i^{\text{tel}}(t)$; record sensitivities alter only this replay input.*

Assumption 2 (Validity of the same-state replay). *After an interval is complete, the replay changes only the admitted request vector. The network topology $\mathcal{G}$, exogenous network data $\mathbf{z}(t)$, and participant set remain fixed. The allocation rule and its parameters also remain fixed. The reactive-power setting is part of $\mathbf{z}(t)$. Available export remains $a_i(t)$ for the replay. For any replayed access $\tilde{\mathbf{x}}(t)$, delivery is $\tilde{y}_i(t) = \min\{\tilde{x}_i(t), a_i(t)\}$. The available-export record must describe the available export in that completed state. The comparison therefore evaluates two allocations against the same available export.*

The second assumption applies when changing the awarded access changes only connection-point delivery and does not alter the underlying available export in the completed interval. An inverter, plant controller, or auditable portfolio process may provide the record. If a different access award would also change underlying available export through inverter control, storage dispatch, flexible demand, or portfolio redispatch, the replay must model those responses. A validated error bound permits the conservative test in (32). Completed delivery also provides a lower bound when an available-export record is missing. ECF does not tighten the later ceiling unless valid records or delivery bounds establish increased delivery in the replay.

Fig. 2 summarizes this timing in Section 2. The model below keeps the completed allocation, post-event replay, and next-

interval ceiling as separate objects.

Let $\mathbf{r}(t) = [r_i(t)]_{i \in \mathcal{N}}$ and $\mathbf{x}^{\mathcal{R}, \mathcal{M}}(t) = [x_i^{\mathcal{R}, \mathcal{M}}(t)]_{i \in \mathcal{N}}$. The vector $\mathbf{r}^{\text{cap}}(t)$ is the request after applying bound C_i . The ECF request ceiling is $\psi_i(t) \in [0, C_i]$. It starts at C_i and is updated in Section 4.4. The admitted request vector is $\mathbf{r}^{\text{adm}}(t) = [r_i^{\text{adm}}(t)]_{i \in \mathcal{N}}$:

$$r_i^{\text{cap}}(t) = \min\{r_i(t), C_i\}, \quad r_i^{\text{adm}}(t) = \min\{r_i^{\text{cap}}(t), \psi_i(t)\}. \quad (2)$$

Before feedback, $\psi_i(t) = C_i$, so the two bounded requests coincide. Every allocation below uses the admitted request, not the unbounded submission.

The vector $\mathbf{z}(t)$ contains exogenous network inputs. These include loads, fixed injections, reactive-power assumptions, network limits, and fixed model coefficients. Voltage, branch flow, and other solved quantities belong to ξ_{net} . They are not components of $\mathbf{z}(t)$. The network feasible set is

$$\mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}(t)) = \left\{ (\mathbf{x}, \xi_{\text{net}}) : \begin{array}{l} \mathbf{f}_{\mathcal{M}}^{\text{eq}}(\mathbf{x}, \xi_{\text{net}}; \mathcal{G}, \mathbf{z}(t)) = \mathbf{0}, \\ \mathbf{f}_{\mathcal{M}}^{\text{ineq}}(\mathbf{x}, \xi_{\text{net}}; \mathcal{G}, \mathbf{z}(t)) \leq \mathbf{0} \end{array} \right\}, \quad (3)$$

where $\mathbf{f}_{\mathcal{M}}^{\text{eq}}$ and $\mathbf{f}_{\mathcal{M}}^{\text{ineq}}$ collect the network equations and limits. The allocation map is

$$\mathbf{x}^{\mathcal{R}, \mathcal{M}}(t) = \Pi_{\mathcal{R}}(\mathbf{r}^{\text{adm}}(t); \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}(t)), \eta_{\mathcal{R}}(t)). \quad (4)$$

The vector $\eta_{\mathcal{R}}(t)$ contains allocation weights and other rule or objective parameters. Network equations and limits instead belong to $\mathcal{F}_{\mathcal{M}}$ through $\mathcal{G}$ and $\mathbf{z}(t)$. ECF changes neither: its ceiling changes the admitted-request input when the ceiling binds. This separation permits the same feedback to be compared under different allocation rules.

The controlled study computes all five allocations with the AC-derived affine second-order cone program (SOCP) proxy in Section 4.3. The closed-form expressions given later for WP, ER, and EA explain local rule behavior; they do not replace the conic network model in the experiments. Radial AC power flow verifies every resulting allocation afterward.

For fixed $\mathcal{G}$, $\mathbf{z}(t)$, $\eta_{\mathcal{R}}(t)$, and current ceilings $\{\psi_i(t)\}_{i \in \mathcal{N}}$, the local request-to-access Jacobian includes the capacity and admission layers:

$$\mathbf{J}_{\mathbf{x}}^{\mathcal{R}, \mathcal{M}}(\mathbf{r}(t); t) = \frac{\partial \mathbf{x}^{\mathcal{R}, \mathcal{M}}(t)}{\partial \mathbf{r}(t)}. \quad (5)$$

For $i \neq j$, a negative off-diagonal entry $\mathbf{J}_{x, j i}^{\mathcal{R}, \mathcal{M}}$ has a direct meaning. Participant j receives less access when r_i increases locally. This derivative is valid only within one smooth regime. If a connection bound, request ceiling, active set, or network constraint changes, we use paired finite differences.

4.2 Request choice and delivery effects elsewhere

The request model explains why a participant can seek more access even when that choice reduces delivery elsewhere. Its one-period payoff values own delivery and charges for unused award. We compare the resulting request with the forecast-based request before introducing the temporal feedback.

4.2.1 Request choice under uncertain available export

Let $a_i^{\text{fcst}}(t)$ be participant i 's private point forecast. The conditional cumulative distribution function $F_i(\cdot | s_i(t))$ describes uncertainty in its available export.

For the payoff derivation, fix the interval, all other requests, the allocation rule, the network model, the exogenous network data, and the conditional available-export CDFs. We suppress those fixed arguments below. The reference request uses the point forecast: $r_i^{\text{ref}}(t) = a_i^{\text{fcst}}(t)$. For participant i , define a larger request by

$$r_i^{\text{chg}}(\gamma) = \gamma r_i^{\text{ref}}(t), \quad \gamma \geq 1. \quad (6)$$

All other requests retain their reference values. We then recompute $\mathbf{x}(\gamma)$ with t , $\mathcal{G}$, $\mathbf{z}(t)$, $\eta_{\mathcal{R}}(t)$, $\mathcal{R}$, and $\mathcal{M}$ fixed. The scalar γ is therefore a request multiplier, not a time argument. The value $\gamma = 1$ gives the reference side of the matched pair.

Let $\vartheta \geq 0$ be the modeled cost per unit of unused award, normalized by the value per unit of delivered export. The realized payoff proxy, measured in MW-equivalent units, is

$$u_i(t) = y_i(t) - \vartheta [x_i(t) - y_i(t)]^+. \quad (7)$$

Here $[\cdot]^+$ is the positive-part operator. For the fixed interval, let $\bar{y}_i(\gamma) = \mathbb{E}[\min\{x_i(\gamma), a_i\} | s_i]$ denote expected delivery. Let $\bar{h}_i(\gamma) = \mathbb{E}[[x_i(\gamma) - a_i]^+ | s_i]$ denote the expected unused award. Expected total delivery is $\bar{Y} = \sum_j \bar{y}_j$. The participant's expected payoff is

$$\begin{aligned} U_i(\gamma; \vartheta) &= \bar{y}_i(\gamma) - \vartheta \bar{h}_i(\gamma) \\ &= \mathbb{E}[\min\{x_i(\gamma), a_i\} - \vartheta [x_i(\gamma) - a_i]^+ | s_i]. \end{aligned} \quad (8)$$

The controlled request model selects the smallest payoff-maximizing multiplier on a fixed grid:

$$\gamma_i^* = \min \left(\arg \max_{\gamma \in \Gamma} U_i(\gamma; \vartheta) \right). \quad (9)$$

Here Γ is the multiplier grid. Let q_{pf} denote the participant reactive-power setting; Q0 and Q95 are specified in Section 4.5. When the experimental factors are shown explicitly, we write $\gamma_i^*(\vartheta, \sigma, \mathcal{R}, q_{\text{pf}})$. The uncertainty scale σ enters through the available-export CDF. Both q_{pf} and $\mathcal{R}$ enter through the allocation response. Participant location and the fixed network model and data remain implicit. For each multi-interval trajectory, this selected multiplier is held fixed, while $a_i^{\text{fcst}}(t)$ supplies the time variation. Choosing the smallest maximizer resolves a payoff tie. The participant uses its private CDF to make this choice. The coordinator receives the submitted request and later the completed-state records. Available export and delivery then obey

$$a_i(t) \sim F_i(\cdot | s_i(t)), \quad y_i(t) = \min\{x_i(t), a_i(t)\}. \quad (10)$$

Equations (8) and (10) use the same available-export CDF. The request is chosen under that model, and delivery is a later realization from it.

4.2.2 Private payoff and effects on aggregate delivery

The following result separates the participant's incentive from the effect on aggregate delivery. We suppress the conditional argu-

ments s_j .

Proposition 4.1 (Marginal effects on private payoff and aggregate delivery). *Fix the other requests, exogenous network data, and conditional available-export CDFs. Suppose $F_j(\cdot | s_j)$ is continuous at x_j and the allocation map is locally differentiable. Write $J_{ji} = \partial x_j / \partial r_i$ for the corresponding entry of the request-to-access Jacobian in (5). With $\bar{Y} = \sum_j \mathbb{E}[\min\{x_j, a_j\} | s_j]$,*

$$\frac{\partial U_i}{\partial r_i} = J_{ii}[1 - F_i(x_i) - \vartheta F_i(x_i)], \quad (11)$$

$$\frac{\partial \bar{Y}}{\partial r_i} = \sum_{j \in \mathcal{N}} J_{ji} [1 - F_j(x_j)]. \quad (12)$$

Proof sketch. For nonnegative available export, $\mathbb{E}[\min\{x, a\}] = \int_0^x [1 - F(u)] du$ and $\mathbb{E}[(x - a)^+] = \int_0^x F(u) du$. Differentiate these two identities and apply the chain rule through J_{ji} . This gives (11)–(12). Section S2 of the *Supporting Information*[†] gives the continuous-regime proof. Section S8 treats request bounds, feedback ceilings, and active-set boundaries. We use derivatives only within one smooth allocation regime. Otherwise, we use a matched finite difference.

If $J_{ii} > 0$, an interior private optimum targets $F_i(x_i) = 1/(1 + \vartheta)$. This choice includes the participant's own delivery and unused award but omits effects on other participants. Those cross-participant effects can be negative when requests share a binding constraint, as shown for the allocation rules in Section 4.3.3. The normalized shortfall cost that makes the private marginal payoff equal the marginal change in aggregate delivery is

$$\vartheta_i^{\text{ext}} = -\frac{\sum_{j \neq i} J_{ji} [1 - F_j(x_j)]}{J_{ii} F_i(x_i)}, \quad J_{ii} F_i(x_i) > 0. \quad (13)$$

The value ϑ_i^{ext} measures the outside delivery effect per unit of marginal unused award. It varies with the allocation state and can change sign. We use it as a local diagnostic.

An allocation response can cross an active-set boundary. The case study therefore also uses a matched finite-difference counterpart. For one participant's request step, define

$$\begin{aligned} \Delta \bar{y}_i &= \bar{y}_i^{\text{chg}} - \bar{y}_i^{\text{ref}}, \\ \Delta \bar{Y}_{\mathcal{N} \setminus \{i\}} &= \sum_{j \neq i} (\bar{y}_j^{\text{chg}} - \bar{y}_j^{\text{ref}}), \\ \Delta \bar{h}_i &= \bar{h}_i^{\text{chg}} - \bar{h}_i^{\text{ref}}. \end{aligned}$$

When $\Delta \bar{h}_i > 0$, the corresponding finite-difference cost ratio is

$$\vartheta_{i,\Delta}^{\text{ext}} = -\frac{\Delta \bar{Y}_{\mathcal{N} \setminus \{i\}}}{\Delta \bar{h}_i}. \quad (14)$$

We report this ratio only when the unused-award increment is positive. A positive value then means that the request step reduces delivery outside the participant. Both (13) and (14) quantify a state-specific delivery effect on other participants.

4.2.3 Matched classification of request effects

Let the nonempty set $\mathcal{K} \subseteq \mathcal{N}$ contain the participants whose requests change jointly. Superscripts ref and chg denote matched

reference and changed-request states. Static cases use the expected changes

$$\begin{aligned} \Delta U_{\mathcal{K}} &= \sum_{i \in \mathcal{K}} (U_i^{\text{chg}} - U_i^{\text{ref}}), \\ \Delta \bar{h}_{\mathcal{K}} &= \sum_{i \in \mathcal{K}} (\bar{h}_i^{\text{chg}} - \bar{h}_i^{\text{ref}}), \quad \Delta \bar{Y} = \bar{Y}^{\text{chg}} - \bar{Y}^{\text{ref}}. \end{aligned} \quad (15)$$

Dynamic intervals use realized payoff and the change in unused award:

$$\begin{aligned} \Delta u_{\mathcal{K}}(t) &= \sum_{i \in \mathcal{K}} [u_i^{\text{chg}}(t) - u_i^{\text{ref}}(t)], \\ \Delta H_{\mathcal{K}}(t) &= \sum_{i \in \mathcal{K}} ([x_i^{\text{chg}}(t) - y_i^{\text{chg}}(t)]^+ - [x_i^{\text{ref}}(t) - y_i^{\text{ref}}(t)]^+). \end{aligned} \quad (16)$$

With tolerance τ_E , a static comparison is classified as an externality when all three components of $(\Delta U_{\mathcal{K}}, \Delta \bar{h}_{\mathcal{K}}, -\Delta \bar{Y})$ exceed τ_E . Thus, the changed-request group's combined expected payoff rises, its combined expected unused award rises, and aggregate delivery falls. A dynamic interval applies the same test to $(\Delta u_{\mathcal{K}}, \Delta H_{\mathcal{K}}, -\Delta Y)$. A productive high request case raises combined payoff without a decrease in aggregate delivery beyond the numerical tolerance. Because $\vartheta \geq 0$, the externality conditions imply that delivery to the changed-request group rises. Aggregate delivery can then fall only if delivery outside that group falls by more. Section S2 of the *Supporting Information*[†] gives the exact bounds. This paired classification is used for mechanism analysis. The completed-state replay in Section 4.4 is the separate operational test that determines whether a later request ceiling changes.

The controlled experiment evaluates (9) on the forecast-linked probability tree in Section 4.6. This gives a one-period request and later available-export realization from the same model; its role is to generate internally consistent controlled cases. Within one smooth allocation regime, a state-specific shortfall cost could match the local marginal delivery effect. The required value changes when the state or active constraints change. ECF instead uses a completed-state replay to decide whether an unused award warrants a later request-ceiling adjustment.

4.2.4 Outcome measures

Every comparison matches the rule, network model, exogenous data, and interval. Participant-group effects are obtained by summing component changes over the named group. Static comparisons use the barred quantities defined above. For a realized matched interval, aggregate metered delivery and unused award are

$$\begin{aligned} Y^{\text{ref}} &= \sum_i y_i^{\text{ref}}, & H^{\text{ref}} &= \sum_i [x_i^{\text{ref}} - y_i^{\text{ref}}]^+, \\ Y^{\text{chg}} &= \sum_i y_i^{\text{chg}}, & H^{\text{chg}} &= \sum_i [x_i^{\text{chg}} - y_i^{\text{chg}}]^+. \end{aligned}$$

Hence $\Delta Y = Y^{\text{chg}} - Y^{\text{ref}}$, and delivery loss is $[-\Delta Y]^+$.

We summarize allocation equality with the capacity-normalized Jain index. Let $\mathbf{C} = (C_i)_{i \in \mathcal{N}}$. All controlled-study connection bounds are positive, and every reported controlled allocation vec-

Table 4 Export-access allocation rules used in the controlled study

Rule	Allocation principle	How requests interact	Experimental role
WP	Common proportional scale ²⁴	Shared scale and binding proxy constraints	One-parameter scaling benchmark
ER	Common absolute reduction ^{27,28}	Shared reduction level	One-parameter reduction benchmark
EA	Common access cap ²⁴	Shared cap and capped set	One-parameter cap benchmark
ME	Maximum total authorized access ^{6,22}	Request bounds and active proxy constraints	Total-access conic rule
FA	Authorized access with a normalized equality term ^{27,29,30}	Request bounds, proxy constraints, and equality term	Access-equality comparison

tor is nonzero. Thus,

$$J_{\text{norm}}(\mathbf{x}, \mathbf{C}) = \frac{(\sum_{i: C_i > 0} x_i / C_i)^2}{|\{i : C_i > 0\}| \sum_{i: C_i > 0} (x_i / C_i)^2}. \quad (17)$$

For a matched pair, $\Delta J_{\text{norm}} = J_{\text{norm}}(\mathbf{x}^{\text{chg}}, \mathbf{C}) - J_{\text{norm}}(\mathbf{x}^{\text{ref}}, \mathbf{C})$. The *Supporting Information*[†] gives the raw Jain index and its boundary conventions. The externality classification uses the delivery, payoff, and unused-award changes defined above; the Jain index is reported separately. The classification therefore differs from ordinary redistribution or forecast error.

4.3 Allocation rules

Whether one request changes access elsewhere depends on how the allocation rule couples participants. The controlled study therefore compares five rules using the same network data, connection bounds, and conic approximation of the network limits. Every resulting allocation then undergoes the same radial AC check for its reactive-power setting. The rules differ in how they select an allocation within that common domain. WP confines the allocation to a common proportional ray. ER and EA use one-parameter families defined by a common reduction and a common cap, respectively. ME and FA optimize directly over the full conic domain. Within each matched comparison, the rule and its formulation are held fixed. The operator $\Pi_{\mathcal{R}}$ maps admitted requests to authorized access. For an optimization-based rule, $W_{\mathcal{R}}$ defines the objective, and $\mathcal{F}_{\mathcal{M}}$ supplies the network constraints.

Table 4 lists the rules. Weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation are abbreviated WP, ER, EA, ME, and FA. The label ME refers here to maximizing authorized access, not metered delivery. Network representation is a separate choice: all five controlled rules use the conic proxy domain for allocation and the same radial AC procedure for subsequent verification.

4.3.1 One-parameter allocation rules

Each rule uses the admitted requests in (2). WP sets $x_i = \rho r_i^{\text{adm}}$ and chooses the largest feasible common scale $\rho \in [0, 1]$. ER sets $x_i = \max\{0, r_i^{\text{adm}} - \lambda\}$ and chooses the smallest feasible common reduction $\lambda \in [0, \max_i r_i^{\text{adm}}]$. EA sets $x_i = \min\{r_i^{\text{adm}}, \mu\}$ and chooses the largest feasible common cap $\mu \in [0, \max_i r_i^{\text{adm}}]$. These

one-parameter allocation families are evaluated against the same conic proxy domain used below; they are not separate physical network models. The *Supporting Information*[†] gives their equations, zero-request boundary cases, and analytical forms for one aggregate allowance. A request change can alter the binding voltage or branch constraint or the membership of a piecewise allocation set. We then re-solve the allocation rather than extend a local formula across that boundary.

4.3.2 Full-domain optimization rules

The following equations describe one fixed interval, so time arguments are suppressed. ME and FA use the same affine conic approximation of the AC network limits without a one-parameter restriction on $\mathbf{x}$. The proxy has three main parts. A second-order cone (SOC) constraint bounds the normalized allocation radius. A cone at each branch end bounds the affine active- and reactive-power prediction. Affine rows bound bus voltage. The model assumes $C_i > 0$ for every participant. With $\mathbf{D}_C = \text{diag}((C_i)_{i \in \mathcal{N}})$, the normalized radius ζ satisfies $\|\mathbf{D}_C^{-1} \mathbf{x}\|_2 \leq \zeta$. Each branch cone has a nonnegative affine right-hand side in ζ . The *Supporting Information*[†] gives the full constraint form and guard-margin definitions. The archived machine-readable records contain the numerical coefficient arrays.

ME Stage 1 is therefore an SOCP. ME Stage 2 and FA contain convex quadratic terms. Introducing epigraph variables for those terms gives equivalent rotated SOC representations. This proxy differs from the branch-flow relaxation in⁴³. Radial AC power flow supplies the physical check. The common primary optimization form is

$$\begin{aligned} \max_{\mathbf{x}, \xi_{\text{net}}} \quad & W_{\mathcal{R}}(\mathbf{x}, \mathbf{r}^{\text{adm}}; \eta_{\mathcal{R}}) \\ \text{s.t.} \quad & 0 \leq x_i \leq r_i^{\text{adm}}, \quad i \in \mathcal{N}, \\ & (\mathbf{x}, \xi_{\text{net}}) \in \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}). \end{aligned} \quad (18)$$

Here $\mathbf{r}^{\text{adm}}$ is the request presented to the optimizer. The vector ξ_{net} contains its endogenous network variables. ME maximizes total authorized access:

$$W_{\text{ME}}(\mathbf{x}) = \sum_i x_i. \quad (19)$$

For FA, the fixed rule-parameter vector is $\eta_{\text{FA}} = (\alpha_{\text{FA}}, \epsilon_{\text{FA}})$. Here $\alpha_{\text{FA}} > 0$ is the MW-valued access-equality weight, and $\epsilon_{\text{FA}} > 0$ is an MW-valued denominator offset. Before FA is solved, the dimensionless ME reference fraction is recomputed from the ME solution for the same admitted request, network model, and exogenous network data:

$$\bar{f}_{\text{ME}}^{\mathcal{M}}(\mathbf{r}^{\text{adm}}) = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \frac{x_i^{\text{ME}, \mathcal{M}}(\mathbf{r}^{\text{adm}})}{r_i^{\text{adm}} + \epsilon_{\text{FA}}}. \quad (20)$$

The FA objective is

$$W_{\text{FA}}(\mathbf{x}, \mathbf{r}^{\text{adm}}; \eta_{\text{FA}}) = \sum_i x_i - \alpha_{\text{FA}} \sum_i \left(\frac{x_i}{r_i^{\text{adm}} + \epsilon_{\text{FA}}} - \bar{f}_{\text{ME}}^{\mathcal{M}}(\mathbf{r}^{\text{adm}}) \right)^2. \quad (21)$$

The quadratic term pulls each participant's request-normalized authorized fraction toward the average fraction in the same-

context ME allocation. It is an allocation-equality term, not a measure of broader social fairness. Table 5 reports the experimental settings.

Let $n = |\mathcal{N}| \geq 2$, and let $\mathbf{1}_n$ be the all-ones vector. ME uses two stages. Stage 1 maximizes total authorized access. Stage 2 applies a fixed, strictly convex tie-break objective while preserving the Stage 1 objective to a certified tolerance. Its feasible set includes

$$\mathbf{1}_n^T \mathbf{x} \geq X_{\min}^{\text{ME}}, \quad X_{\min}^{\text{ME}} = X_1^{\text{ub}} - \tau_H. \quad (22)$$

Here X_1^{ub} is a valid Stage 1 upper bound, and $\tau_H > 0$ is the hierarchy tolerance. Under the certificate conditions in the *Supporting Information*[†], the selected allocation is unique and its total authorized access is within τ_H of the Stage 1 optimum.

FA uses the same-context ME solution only to compute (20); it does not impose the ME near-optimality floor. It can therefore trade total authorized access against request-normalized access equality as α_{FA} changes. The *Supporting Information*[†] gives the complete ME hierarchy, its unique-selection and authorized-access bounds, the cone formulation, and the KKT sensitivity derivation. Deterministic selection ensures that a paired response reflects changed requests rather than an arbitrary choice among ME optima. At an active-set change, we re-solve both allocations and use their paired difference. The conic and AC checks retain the distinct roles described above.

4.3.3 How one request changes access elsewhere

The allocation rule determines how one request changes access at other connections. WP couples requests through a common scale. ER applies a common reduction to admitted requests, and EA applies a common access cap. Under ME and FA, a request change can also change which network constraints are active.

For the local derivations, fix the interval, rule, network model, exogenous data, and rule parameters. Requests also remain below their fixed upper bounds and feedback ceilings. Thus, $r_i^{\text{adm}} = r_i$ locally. A binding ceiling ends this smooth regime. We write $r_{i_0} = r_{i_0}^{\text{ref}}(t)$ and suppress fixed indices. Consistent with (6), $r_{i_0}^{\text{chg}}(\gamma_{i_0}) = \gamma_{i_0} r_{i_0}$. The expression $x_i(\gamma_{i_0})$ denotes access after applying this multiplier. It is not a time argument.

Paired comparison procedure. For each comparison, we fix the interval, exogenous data, rule, and network model. We also fix the available-export input appropriate to the analysis: the conditional CDF in an expected comparison or the realized available-export record in a completed-interval comparison. We solve the reference and changed requests without altering any other input. In the controlled comparisons, the changed set $\mathcal{K}$ contains one or five participants, each with its own multiplier. We compute participant-level access and delivery differences before aggregation. We use a local sensitivity only while the active regime is unchanged; otherwise, we use the matched finite difference. The reference and changed allocations undergo the same radial AC power-flow check.

Definition 4.1 (Outside access loss after a request increase). *With the quantities above fixed, define $\Delta x_j(\gamma_{i_0}) = x_j(\gamma_{i_0}) - x_j(1)$. Participant $j \neq i_0$ experiences request-induced access loss when $\Delta x_j(\gamma_{i_0}) < 0$. This change occurs before delivery is observed. The*

vector $\Delta \mathbf{x}(\gamma_{i_0})$ retains all signed access changes, including any increases elsewhere.

WP provides a transparent algebraic illustration. For this illustration, let $h_{m,i}$ be nonnegative coefficients of linear allowance row m , and let B_m be its positive allowance. The resulting common scale is

$$\rho = \min\{1, \min_{m: D_m > 0} B_m / D_m\}, \quad D_m = \sum_i h_{m,i} r_i.$$

When one such row binds, increasing r_{i_0} weakly lowers ρ . Access can then fall for other participants. The final allocation still satisfies the modeled limit because every request enters the same scaling denominator. The controlled experiment instead determines the common scale against the full conic proxy domain.

ER and EA responses also depend on the active participant set. The following local expressions explain that dependence; Section S6 of the *Supporting Information*[†] gives the full cases. They illustrate the rule coupling rather than replace the conic allocation used in the experiments.

Proposition 4.2 (WP outside-access response under a linear allowance). *Let m^* be the only active constraint near the reference point. Set $B = B_{m^*} > 0$ and $h_i = h_{m^*,i} \geq 0$. Let $D_{\text{oth}} = \sum_{i \neq i_0} h_i r_i$. If the active constraint is unchanged, then, for $j \neq i_0$,*

$$\begin{aligned} \Delta x_j(\gamma_{i_0}) &= - \frac{B r_j h_{i_0} r_{i_0} (\gamma_{i_0} - 1)}{(D_{\text{oth}} + h_{i_0} \gamma_{i_0} r_{i_0})(D_{\text{oth}} + h_{i_0} r_{i_0})}, \\ \frac{\partial x_j}{\partial r_{i_0}} &= - \frac{B r_j h_{i_0}}{(\sum_i h_i r_i)^2}. \end{aligned} \quad (23)$$

Thus, a larger request weakly lowers other allocations through the common scale. The modeled constraint remains satisfied. In the binding smooth regime, the decrease is strict when $\gamma_{i_0} > 1$, $r_j > 0$, $r_{i_0} > 0$, and $h_{i_0} > 0$; otherwise, it can be zero. Substitute $x_j = B r_j / (D_{\text{oth}} + h_{i_0} r_{i_0})$ to obtain the difference and derivative.

For ER, fix a nonempty positive-allocation set $\mathcal{A}_+$ that contains i_0 . A change in the common reduction λ changes every other positive allocation by the same amount. The change in λ depends on the active proxy constraint. Under the special single equal-weight aggregate allowance $\sum_i x_i \leq B$, a request increase raises λ by $(\gamma_{i_0} - 1) r_{i_0} / |\mathcal{A}_+|$ while the same set remains active.

For EA, let $\mathcal{C}$ be the nonempty capped set and $\mathcal{U} = \mathcal{N} \setminus \mathcal{C}$ the uncapped set. The common cap μ is likewise determined by the active proxy constraint. Under the same single equal-weight aggregate allowance, a request increase from $i_0 \in \mathcal{U}$ lowers μ by $(\gamma_{i_0} - 1) r_{i_0} / |\mathcal{C}|$. A request already above the cap is locally inactive. ER and EA are piecewise linear in their common scalar, but the request-to-scalar response is network dependent in the full conic model. The *Supporting Information*[†] gives the special fixed-membership formulas. A binding ECF ceiling makes a further increase in the submitted request locally ineffective. We evaluate active-set and boundary changes with paired finite differences.

4.4 ECF replay, ceiling update, and comparison controls

Under-delivery alone does not justify a later restriction. Network access may be slack, or no other participant may be able to use

it. ECF therefore combines metered delivery with the available-export record in Assumption 1. The unused-award group is

$$\mathcal{N}_{\text{ua}}(t) = \{i : x_i(t) - y_i(t) > \varepsilon_x\}, \quad (24)$$

where $\varepsilon_x > 0$ is the unused-award tolerance. The replay retains the admitted request of every participant outside this group. It caps only a group member's replay request at metered delivery:

$$r_i^{\text{rpl}}(t) = \begin{cases} \min\{r_i^{\text{adm}}(t), y_i(t)\}, & i \in \mathcal{N}_{\text{ua}}(t), \\ r_i^{\text{adm}}(t), & i \notin \mathcal{N}_{\text{ua}}(t). \end{cases} \quad (25)$$

The completed allocation remains unchanged. Under Assumption 2, the replay uses the same rule and network model. Its delivery uses the specified available-export record:

$$\mathbf{x}^{\text{rpl}}(t) = \Pi_{\mathcal{R}}\left(\mathbf{r}^{\text{rpl}}(t); \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}(t)), \eta_{\mathcal{R}}(t)\right), \quad (26)$$

$$y_i^{\text{rpl}}(t) = \min\{x_i^{\text{rpl}}(t), a_i^{\text{tel}}(t)\}. \quad (27)$$

Let

$$\Delta X_{\text{out}}^{+, \text{rpl}}(t) = \sum_{j \notin \mathcal{N}_{\text{ua}}(t)} [x_j^{\text{rpl}}(t) - x_j(t)]^+, \quad (28)$$

$$\Delta Y_{\text{out}}^{+, \text{rpl}}(t) = \sum_{j \notin \mathcal{N}_{\text{ua}}(t)} [y_j^{\text{rpl}}(t) - y_j(t)]^+, \quad (29)$$

$$\Delta Y_{\text{sys}}^{\text{rpl}}(t) = \sum_j y_j^{\text{rpl}}(t) - \sum_j y_j(t). \quad (30)$$

The first two quantities sum positive participant-level changes outside the unused-award group. The third is the signed change in aggregate participant delivery. A replay request inside the unused-award group does not exceed that participant's metered delivery, so its replay delivery cannot exceed its completed delivery. A positive aggregate change therefore also implies a positive signed net delivery change outside the group; the positive-part sum identifies delivery received by outside participants who gain. The binary indicator $g(t)$ records whether the replay meets all four conditions. We use the positive tolerance τ_{rpl} :

$$g(t) = \begin{cases} 1, & \begin{aligned} &\text{the replay allocation is obtained,} \\ &\text{and the specified AC checks pass,} \\ &\Delta X_{\text{out}}^{+, \text{rpl}}(t) > \tau_{\text{rpl}}, \\ &\Delta Y_{\text{out}}^{+, \text{rpl}}(t) > \tau_{\text{rpl}}, \\ &\Delta Y_{\text{sys}}^{\text{rpl}}(t) > \tau_{\text{rpl}} \end{aligned} \\ 0, & \text{otherwise.} \end{cases} \quad (31)$$

The required AC checks depend on the network representation. The controlled study checks convergence, power balance, voltage, and apparent power at both branch ends for allocation and delivery states (Section 4.5). The Australian study checks load-only and authorized-export endpoints as well as profile-derived delivery, using the customer-voltage and native transformer limits in Section 4.7. Both studies evaluate the replay delivery state separately from the awarded-access state. The test evaluates the completed unused-award group as a whole. It does not assign

responsibility to one member. The indicator remains zero if any condition fails: no outside access is released, outside or total delivery does not rise, or the replay fails the specified AC checks. When $g(t) = 1$, the later target applies to every member of the unused-award group.

The controlled study sets $a_i^{\text{tel}}(t) = a_i(t)$. Suppose a deployment instead certifies the error bound $|a_i^{\text{tel}}(t) - a_i(t)| \leq \delta_i$. Define the lower record

$$a_i^{\text{L}}(t) = \max\{y_i(t), a_i^{\text{tel}}(t) - \delta_i\}, \quad y_i^{\text{rpl,L}}(t) = \min\{x_i^{\text{rpl}}(t), a_i^{\text{L}}(t)\}. \quad (32)$$

For operational use, the error radius $\delta_i \geq 0$ must come from independent validation data. The numerical comparison in Section S10.9 of the *Supporting Information*[†] instead uses prescribed radii to examine the resulting delivery tradeoff. The vectors $\mathbf{a}(t)$ and $\mathbf{y}^{\text{rpl,L}}(t)$ collect the corresponding participant-level quantities.

Proposition 4.3 (Conservative delivery recovery under bounded record error). *Suppose metered delivery is exact and $y_i(t) \leq a_i(t)$. Also suppose $|a_i^{\text{tel}}(t) - a_i(t)| \leq \delta_i$. Evaluate the two delivery-recovery conditions in (31) with $\mathbf{y}^{\text{rpl,L}}(t)$. If both exceed τ_{rpl} , they also exceed that threshold under the componentwise true replay delivery $\min\{\mathbf{x}^{\text{rpl}}(t), \mathbf{a}(t)\}$.*

Proof sketch. The error bound gives $y_i(t) \leq a_i^{\text{L}}(t) \leq a_i(t)$. Therefore, $y_i^{\text{rpl,L}}(t) \leq \min\{x_i^{\text{rpl}}(t), a_i(t)\}$ componentwise. Outside delivery recovery and total delivery recovery both increase with replay delivery. Section S3 of the *Supporting Information*[†] gives the complete proof. The AC checks are not implied by this bound and must still be performed. In a deployment, the error radius must be established before applying the replay test.

Recall that $\psi_i(t) \in [0, C_i]$ is the ECF request ceiling in MW. The vector $\boldsymbol{\psi}(t) = (\psi_i(t))_{i \in \mathcal{N}}$ collects all ceilings. Each starts at $\psi_i(1) = C_i$. The post-event target is

$$\psi_i^{\text{tar}}(t) = \begin{cases} y_i(t), & g(t) = 1 \text{ and } i \in \mathcal{N}_{\text{ua}}(t), \\ C_i, & \text{otherwise,} \end{cases} \quad (33)$$

and the one-step update is

$$\begin{aligned} \psi_i(t+1) &= (1 - \beta_{\text{ECF}})\psi_i(t) + \beta_{\text{ECF}}\psi_i^{\text{tar}}(t), \\ 0 &\leq \beta_{\text{ECF}} \leq 1. \end{aligned} \quad (34)$$

The next request first meets the fixed bound C_i . It then meets the ECF ceiling:

$$r_i^{\text{cap}}(t+1) = \min\{r_i(t+1), C_i\}, \quad (35)$$

$$r_i^{\text{adm}}(t+1) = \min\{r_i^{\text{cap}}(t+1), \psi_i(t+1)\}. \quad (36)$$

For fixed $\psi_i(t+1)$, a larger raw request cannot bypass a binding ceiling. During consecutive intervals that do not meet the replay conditions, the ceiling recovers geometrically toward C_i for $0 < \beta_{\text{ECF}} \leq 1$. At $\beta_{\text{ECF}} = 0$, the ceiling does not change. A participant outside the unused-award group receives no new restriction from that interval. The update also requires no estimate of the private available-export CDF.

The shortfall-cost benchmark changes request choice through (8). It has no post-event request-ceiling state. Mismatch-only

feedback instead smooths each delivery-to-award ratio with gain β_{ratio} and multiplies the resulting state by C_i to obtain the next ceiling. For an award above ϵ_x , a shortfall makes the target ratio less than one. The ceiling falls only if that target is below the previous state; it can rise when the ratio improves. Full delivery sets the target to one. This update uses no replay. Section S3 of the *Supporting Information*[†] gives the recursion and initialization.

The equal-total control isolates where ECF reduces requests. Let $R_{\text{cap}}(t) = \sum_i r_i^{\text{cap}}(t)$. If $r_i^{\text{adm,ECF}}(t)$ is the admitted request under ECF, define

$$\Delta R_{\text{ECF}}(t) = \sum_i \left[r_i^{\text{cap}}(t) - r_i^{\text{adm,ECF}}(t) \right]. \quad (37)$$

When $R_{\text{cap}}(t) > 0$, define

$$v(t) = 1 - \frac{\Delta R_{\text{ECF}}(t)}{R_{\text{cap}}(t)}, \quad r_i^{\text{adm,uni}}(t) = v(t) r_i^{\text{cap}}(t). \quad (38)$$

The superscript uni denotes the uniform scaling used by this equal-total control. Set $v(t) = 1$ when $R_{\text{cap}}(t) = 0$. ECF and this comparison remove exactly the same total request in every interval:

$$\sum_i \left[r_i^{\text{cap}}(t) - r_i^{\text{adm,uni}}(t) \right] = \Delta R_{\text{ECF}}(t). \quad (39)$$

They differ only in which participants' admitted requests are reduced. For the following interval-level identity, time arguments are suppressed. Because $Y = X - H$, where $X = \sum_i x_i$ and $H = \sum_i (x_i - y_i)$, their delivery difference is

$$Y^{\text{ECF}} - Y^{\text{uni}} = (X^{\text{ECF}} - X^{\text{uni}}) + (H^{\text{uni}} - H^{\text{ECF}}). \quad (40)$$

This identity shows why reducing total authorization is neither necessary nor sufficient to improve delivery. Less authorization helps only if its unused portion falls by more; additional authorization helps when it exceeds any increase in its unused portion. Equal request totals need not produce equal award totals because the spatial request distribution affects the binding network constraints. Section S4 of the *Supporting Information*[†] derives these conditions and a sufficient condition for a feasible redistribution to increase delivery. They describe the resulting allocations; a later update must still be tested under the conditions of the later interval. ECF changes a later request ceiling and thereby, when binding, the request input; it does not change the network feasible set. Section S5 of the *Supporting Information*[†] separately examines retained network margins, which alter a network allowance rather than a request ceiling.

4.5 Controlled medium-voltage network and AC checks

The controlled study tests allocation behavior inside a fixed medium-voltage operating envelope. It uses the 141-bus radial feeder in⁴⁴. The network data, participant locations, and synthetic branch ratings used for the two-ended checks are fixed before the request comparisons. Table 5 reports the numerical case settings. The participant reactive-power setting is $q_{\text{pf}} \in \{\text{Q0}, \text{Q95}\}$. Q0 sets participant reactive power to zero. Q95 uses 0.95 power factor with reactive-power absorption.

Each participant represents an export connection point or ag-

gregated portfolio on the medium-voltage feeder. This controlled representation isolates the request-allocation-delivery mechanism. The public-data study separately examines service connections on published medium- and low-voltage models.

The conic proxy defines the allocation domain through affine voltage rows and two-ended branch-flow cones, given in Section S9 of the *Supporting Information*[†]. A separate nonlinear radial alternating-current (AC) power-flow calculation checks every reference, changed-request, and replay allocation under the same reactive-power setting. Distinct observed-delivery and replay-delivery vectors are also applied as injections to the same AC model. A matched pair or ECF interval is physically feasible only when every applicable award and delivery state passes.

For each interval, let $\phi_{\text{pf}} = 0$ in Q0 and $\phi_{\text{pf}} = \arccos(0.95)$ in Q95. Let $P_{D,b}(t)$ and $Q_{D,b}(t)$ denote the active and reactive demand at bus b before participant export; positive values denote demand. Positive $x_i(t)$ denotes awarded participant export. Because Q95 represents reactive-power absorption, an award state is mapped to net bus demand as

$$\begin{aligned} P_{D,b}^{\text{net}}(t) &= P_{D,b}(t) - \sum_{i:v(i)=b} x_i(t), \\ Q_{D,b}^{\text{net}}(t) &= Q_{D,b}(t) + \sum_{i:v(i)=b} x_i(t) \tan \phi_{\text{pf}}. \end{aligned} \quad (41)$$

For an observed- or replay-delivery check, the same map is evaluated with the corresponding delivery vector in place of $\mathbf{x}(t)$. Let $\mathcal{V}_b(t)$ be the nonlinear AC bus-voltage phasor. Its per-unit magnitude is $v_b(t) = |\mathcal{V}_b(t)|$, with limits $V_{\min} \leq v_b(t) \leq V_{\max}$. For branch ℓ , the complex power at reporting end $e \in \{\text{fr}, \text{to}\}$ is $S_{\ell,e}(t) = P_{\ell,e}(t) + jQ_{\ell,e}(t)$; its magnitude is the apparent power. Every monitored branch has a fixed positive synthetic scenario rating $\bar{S}_{\ell}^{\text{syn}}$. Its two-ended loading ratio is

$$\text{LR}_{\ell}(t) = \frac{\max_{e \in \{\text{fr}, \text{to}\}} |S_{\ell,e}(t)|}{\bar{S}_{\ell}^{\text{syn}}}. \quad (42)$$

Each AC calculation must converge and satisfy active- and reactive-power balance, the voltage limits, and $\text{LR}_{\ell}(t) \leq 1$ for every monitored branch. A matched pair passes only when both sides satisfy these conditions at the same reactive-power setting. Rooted branch orientation is used only to report flow direction; thermal eligibility depends on the apparent-power magnitude at both ends. The fixed scenario ratings define the controlled operating envelope. They are not native equipment ratings for the 141-bus feeder.

4.6 Controlled multi-interval experiment

The static experiment distinguishes productive larger requests from requests that reduce total delivery. The multi-interval experiment then tests whether ECF increases later delivery after an unused-award episode meets the four replay conditions. Within each matched comparison, we hold the allocation rule, exogenous network data, reactive-power setting, and AC power-flow criteria fixed. The study combines the network in Section 4.5 with the five rules in Table 4. Section S3 of the *Supporting Information*[†] lists the temporal sequences and the complete comparison design.

One conditional available-export tree links request choice to realized delivery:

$$a_i^{(k)}(t) = \min\left\{C_i, (1 + \sigma \zeta_k) a_i^{\text{fcst}}(t)\right\}, \quad (43)$$

$$(\zeta_k)_{k=1}^5 = (-2, -1, 0, 1, 2), \quad (\pi_k)_{k=1}^5 = \frac{1}{16}(1, 4, 6, 4, 1).$$

The standardized support $(\zeta_k)_{k=1}^5$ has weighted mean zero and weighted variance one. The tested values of the dimensionless uncertainty scale σ keep $1 + \sigma \zeta_k$ nonnegative. Node k occurs with probability π_k , and the upper bound C_i clips its available export. Thus, the participant's forecast anchors both the ex-ante request choice and the later available-export realization. Equation (9) selects the request multiplier from the grid in Table 5.

The static experiment spans 30 participant locations, five allocation rules, both reactive-power settings, five uncertainty levels, and nine normalized shortfall-cost levels. The resulting 13,500 request-choice cases evaluate the smallest payoff-maximizing multiplier and support the matched classification of request increases before any temporal feedback is applied.

The multi-interval experiment carries the selected γ_i^* from the corresponding static setting into a 16-interval trajectory. Each participant in the changed-request set $\mathcal{K}$ submits

$$r_i^*(t) = \gamma_i^*(\vartheta, \sigma, \mathcal{R}, q_{\text{pf}}) a_i^{\text{fcst}}(t), \quad (44)$$

before available export is realized; participants outside $\mathcal{K}$ submit their forecast requests. Three profiles contain the same support-node counts and differ only in temporal order: clustered, intermittent, and shock-recovery. Cyclic participant offsets prevent all participants from realizing the same node at once. Because this is a controlled experiment, the post-event record equals the simulated available export, $\mathbf{a}^{\text{tel}}(t) = \mathbf{a}(t)$.

The spatial cases change one participant, five electrically local participants, or five dispersed participants. Every changed participant uses the multiplier that maximizes its expected payoff in the corresponding single-participant, single-interval calculation. The five-participant cases therefore apply heterogeneous request changes simultaneously. We compare the shortfall-cost benchmark, mismatch-only feedback, three ECF gains, and the equal-total control. All 1,620 trajectories are matched by allocation rule, temporal profile, changed-request set, normalized shortfall cost, and reactive-power setting. Their 25,920 interval comparisons preserve the same matching.

Table 5 summarizes the controlled design; Section S3 of the *Supporting Information*[†] gives the full trajectory construction and comparison counts.

4.7 Australian public-data replay

The public-data replay combines two Australian sources. The first is a published unbalanced OpenDSS model derived from an 11-kV feeder⁴⁵. It contains 1,255 service connections and 33 low-voltage sections, of which 30 carry mapped service loads. Its topology and equipment data come from geographic information system records and documented engineering completions.

The chronological source is the Ausgrid panel of 300 gross-

Table 5 Controlled medium-voltage multi-interval design

Factor	Tested levels
Allocation rule	WP, ER, EA, ME, FA
Controlled feeder	141-bus radial feeder; demand scale 0.70; 30 fixed participant buses
Reactive-power setting q_{pf}	Q0, Q95
AC limits	$0.95 \leq v_b \leq 1.05$ pu; $\text{LR}_\ell \leq 1$ using both branch ends
Request-choice grid Γ	$\{1.0, 1.1, \dots, 2.0\}$; smallest maximizer
Static uncertainty σ	0, 0.05, 0.10, 0.20, 0.30
Static normalized shortfall cost ϑ	0, 0.05, 0.10, 0.25, 0.50, 1, 2, 5, 10
Dynamic (σ, ϑ)	$\sigma = 0.20$; $\vartheta \in \{0.1, 10\}$
Temporal profile	Clustered, intermittent, shock-recovery
Changed-request set	One participant; local five; dispersed five
Comparison mechanism	Shortfall-cost benchmark; mismatch-only feedback $\beta_{\text{ratio}} = 0.5$; ECF $\beta_{\text{ECF}} = 0.25, 0.5, 1$; equal-total control
FA parameters	$\alpha_{\text{FA}} = 0.20$ MW; $\epsilon_{\text{FA}} = 10^{-6}$ MW; α_{FA} sensitivity: 0.05, 0.10, 0.20, 0.40, 0.80 MW
Controlled-study decision tolerances	$\epsilon_k = 10^{-9}$ MW; $\tau_{\text{pl}} = \tau_E = 2 \times 10^{-10}$ MW
Horizon	16 intervals with exact tree frequencies
Sample size	1,620 trajectories; 25,920 matched interval comparisons, each AC-checked

metered solar homes⁴⁶. It provides half-hour gross PV generation, general consumption, and controlled-load energy from July 2010 to June 2013. The profile and network sources have no common customer identifier. We therefore use one deterministic profile-to-service-point mapping. All 300 profiles are used before any profile is reused. This primary mapping remains fixed within all primary trajectories. A separate sensitivity uses five alternative placements of the same profiles, as specified below.

The three source years serve separate roles. Data from 2010–2011 estimate a half-hour-specific forecast increment from day-to-day changes. Data from 2011–2012 select the feeder load scale and qualify the allocation-stage upper-voltage ceiling. Data from 2012–2013 provide the later assessment trajectories. Let d index calendar days, and let $\tau \in \{1, \dots, 48\}$ index half-hours. We divide each half-hour energy value by 0.5 h and 1,000 to obtain average power in MW.

The primary analysis retains source rows marked as actual. Estimated or substituted gross generation and general consumption are excluded. Controlled load is required whenever that customer-year contains the channel. We do not impute missing values.

The request construction uses the previous calendar day. A network day is eligible only when every distinct source profile used by the fixed mapping is complete on both the current and previous calendar day and both dates pass the chronology check. Reusing a source profile at several service points does not repeat its completeness requirement. We also exclude Australian daylight-saving transition dates from the state recursion. The assessment then contains 80 network-day trajectories, or 3,840 half-hours per control. It includes 21 days in January, 11 in April, and 24 each in May and June. No other assessment month contains an eligible day. A complete-customer sensitivity retains customers with records in all three years under the same loading assumptions.

The forward-uptake scenario assigns export participation to 418 of the 1,255 service connections. We scale the 300 PV profiles to a mean capacity of 9.9 kWp. Fixed profile reuse across 418 participants gives a participant mean of 9.784 kWp. The 9.9-kWp scale uses the reported mean Australian rooftop installation in the March quarter of 2025⁴⁷. This defines a forward uptake case rather than a historical feeder reconstruction.

Let $\iota(i)$ denote the fixed profile mapping, and let λ_{PV} and λ_L

denote the fixed PV and load scales. After conversion to MW, GG, GC, and CL denote gross PV generation, general consumption, and controlled load, respectively. The scenario's profile-derived available-export proxy is

$$a_i^{\text{prof}}(d, \tau) = \left[\lambda_{\text{PV}} \text{GG}_{i(i)}(d, \tau) - \lambda_L \left(\text{GC}_{i(i)}(d, \tau) + \text{CL}_{i(i)}(d, \tau) \right) \right]^+. \quad (45)$$

For control c , modeled connection-point delivery is

$$y_i^c(d, \tau) = \min\{x_i^c(d, \tau), a_i^{\text{prof}}(d, \tau)\}. \quad (46)$$

The primary replay record is $a_i^{\text{tel}}(d, \tau) = a_i^{\text{prof}}(d, \tau)$; record sensitivities change only a_i^{tel} . The profile-derived proxy represents PV-minus-load surplus before network access is applied, so it can exceed C_i . The bound limits the admitted request and export award, and hence $y_i^c \leq x_i^c \leq C_i$; it does not clip the underlying proxy.

At each half-hour, the raw request begins with the previous day's a_i^{prof} at the same half-hour. It adds the positive part of the 90th percentile of same-half-hour day-to-day changes estimated from 2010–2011, and then meets the fixed connection bound C_i . This constructed forecast replaces the single-interval request-choice model used in the controlled study. Household demand uses 0.95 lagging power factor, whereas PV export uses unity power factor. For allocation, this retrospective replay treats contemporaneous half-hour household demand as fixed exogenous data; load-forecast error is not modeled.

The Australian replay sets $\varepsilon_x^{\text{AU}} = \tau_{\text{rpl}}^{\text{AU}} = 2 \times 10^{-6}$ MW (2 W). These are numerical thresholds for the unused-award and recovery tests, not estimates of measurement accuracy. The controlled thresholds in Table 5 are smaller and serve its deterministic calculations. Record uncertainty is treated separately through δ_i or the stated sensitivity transformations. Section S3 of the *Supporting Information*[†] places both studies' thresholds side by side.

At each half-hour, an unbalanced nonlinear AC search scales the admitted request as $x_i(d, \tau) = \rho r_i^{\text{adm}}(d, \tau)$, with $\rho \in [0, 1]$. If the full request fails, zero-anchored bisection yields the largest feasible lower-bracket factor encountered. This is a feasible-factor search; global maximality would additionally require monotonic feasibility or a search of the full factor interval. It stops when the factor bracket is at most 10^{-5} or after 24 iterations. For every candidate, the load-only and authorized-export endpoints must converge, remain above 0.90 pu, and satisfy native transformer nameplates; the authorized endpoint must also remain below the 1.07-pu allocation-stage upper-voltage ceiling. The profile-derived delivery state is screened against the 0.90–1.10-pu physical voltage range and the same transformer limits. Because the public model provides no usable native line ampacities, the physical scope covers customer voltage and transformer loading. Customer voltage is measured across each modeled service terminal: phase to neutral for wye connections and phase to phase for delta connections. The nonlinear solution tolerance is 10^{-10} .

The scenario examines allocation of constrained export access under feasible operating conditions. It uses a load multiplier of 0.30 and a 1.07-pu upper-voltage ceiling during authorization, retaining 0.03 pu below the 1.10-pu physical upper limit. Feeder-setting comparisons use no-feedback operation to check physical

feasibility, retain useful delivery, and preserve intervals in which network limits restrict requests. The selected settings apply to all controls, sensitivities, nested sections, and representative topologies. Delivery states are checked separately because an authorization margin alone does not guarantee their feasibility.

In the primary comparison, each assessed day starts with $\psi_i(1) = C_i$. The controls are no feedback, previous-day persistence, ECF with $\beta_{\text{ECF}} = 0.5$, and the equal-total control. Previous-day persistence uses the preceding day's same-half-hour available-export proxy without the added forecast increment or any replay. These comparisons retain state memory within each day.

The paired comparisons separate update qualification, memory, and information quality. ECF without replay qualification retains the update formula and gain 0.5 but moves ceilings toward completed delivery after every unused award. Each control evolves its own ceilings and deliveries under common raw requests and network conditions (SI S10.6). The continuous-state comparison instead carries ceilings across midnight within the ten observed blocks of 1–35 days. It retains the three tested gains and starts a new block at capacity only after a calendar gap (SI S10.8).

The record comparisons retain unperturbed availability for delivery assessment. One set scales the post-event record by 0.9 or 1.1, delays it by one half-hour, or replaces a deterministic 10% of participant records with completed delivery. All transformed records are floored at current modeled delivery; the first lagged half-hour also uses delivery alone. Each setting generates its own equal-total control. A second set compares point-record and lower-bound recovery tests at error radii of 1%, 5%, and 10% of registered capacity, under common positive and negative biases. These controls evolve separately. Reference availability checks their recovery decisions afterward; it does not select their updates. The exact-record case and six bias settings per treatment give 13 settings (SI S10.9). These are prescribed error tests, not a calibration of instrument accuracy.

Gain sensitivities use $\beta_{\text{ECF}} \in \{0.25, 1.00\}$ around the primary 0.5. The complete-customer cohort keeps the 0.30 load multiplier while changing customer composition and the eligible calendar (SI S10.7). A separate behavioral comparison lets participants revise request-choice probabilities from their own completed delivery, carrying preferences and ceilings across consecutive dates. Its fixed-probability comparison uses the same candidate requests and random draws. Independently evolving controls and the conditional equal-total shadow are distinguished in SI S10.15.

Spatial comparisons change profile placement, PV capacity, or network structure separately. Five mappings reassign whole profiles within participant and nonparticipant groups, preserving profile reuse, PV capacities and forecast histories, participant locations, phase connections, and aggregate inputs (SI S10.10). The capacity comparison keeps the original mapping and all 418 participant locations, scales gross PV and registered capacities by 0.50, 0.75, 1.00, 1.25, or 1.50, and recalculates net available export and forecast increments under the same construction rule (SI S10.11). These comparisons retain the primary dates, load scale, network limits, gain, and initialization.

The network panels use 13 published LV sections of the parent

collection and nine representative Australian LV models⁴². Each section has at least 20 mapped connections and five export participants. Representative-model inclusion requires a convergent service-voltage check with household injections set to zero, a usable native transformer nameplate, and the specified minimum connection counts. Original phase connections are retained. All panels apply the common chronology, mapping rule, controls, and network-setting policy; SI S10.12–S10.13 report each network separately.

Energy is the sum of half-hour average power multiplied by 0.5 h. The statistical unit is the paired network-day trajectory. Intervals, participants, and nested LV sections remain observations within that trajectory. We report paired-day bootstrap percentile intervals and dependent wild-bootstrap intervals. The latter preserve short-range calendar dependence while retaining all eligible days. Section S10 of the *Supporting Information*[†] gives the filtering, scenario construction, energy identities, sensitivities, and topology results.

For control c , let E_X^c denote authorized export energy over the eligible panel, and let E_Y^c denote delivered export energy. Section S10 gives their half-hour sums and the unused-authorization identity.

The same nonlinear solves record source-terminal active power and circuit loss. This shows how a change in connection-point delivery affects upstream exchange. Under the OpenDSS convention, negative source power supplies the feeder. Positive power is export toward the upstream system⁴⁸. Let E_{up}^c be the half-hour source-energy sum for control c . A positive paired change means less upstream import or more upstream export. Circuit loss is reported separately. Because the model contains voltage-dependent passive elements, reported circuit loss need not equal the entire difference between connection-point delivery and source-terminal exchange.

5 Conclusions

An unused export award does not, by itself, establish that another participant could have delivered more electricity. Larger requests can also protect useful delivery when output is uncertain. ECF therefore tests whether an alternative allocation of a completed interval would have enabled more delivery elsewhere and in total before adjusting later request ceilings.

The results show that reducing the same total request can produce different delivery outcomes depending on whose requests are reduced. Network location and available output determine whether released access can be used. Less unused authorization is not, by itself, a sufficient measure of improvement. Selective updates can increase delivered electricity by changing the distribution of access, without requiring a reduction in total authorization.

Finding that others could have delivered more in a completed interval does not determine how strongly future access should change. An update that follows the latest shortfall too closely can restrict export when output recovers. Effective feedback must therefore combine a justified reason to adjust requests with an update that preserves access for later recovery.

Completed operating records can thus support better use of

existing export capacity while retaining the network limits, allocation rule and settlement arrangements. The objective is to deliver more available electricity, rather than minimize unused permission. Field evaluation should validate the available-export records and examine how participants alter requests when they anticipate future ceiling adjustments.

Author contributions

Junjie Huang: conceptualization, methodology, software, formal analysis, investigation, visualization, data curation, and writing—original draft. Tao Yu and Yusheng Xue: conceptualization, supervision, funding acquisition, and writing—review and editing. Zhenning Pan, Yufeng Wu, and Wencong Xiao: methodology, validation, and writing—review and editing. Rankai Zhu: validation, data curation, and writing—review and editing. Jie Huang: validation and writing—review and editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The SI, simulation and analysis code, calculated result tables, figure data, and computational environment specification are deposited in the project repository[†]. The repository is currently private for author review; access for editors and reviewers will be arranged before submission. The cited 141-bus case, CSIRO/GridQube network collection (DOI 10.25919/ghnz-bk28), and Ausgrid Solar Home Electricity Data are obtained from their original distributors under the applicable terms. The repository provides source identifiers, preprocessing code, parameter files, date selections, mapping tables, file hashes, and reproduction instructions. It distinguishes checks of saved results from full simulation reruns and identifies the external inputs and auxiliary caches needed for the latter.

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